

BFS Traversal

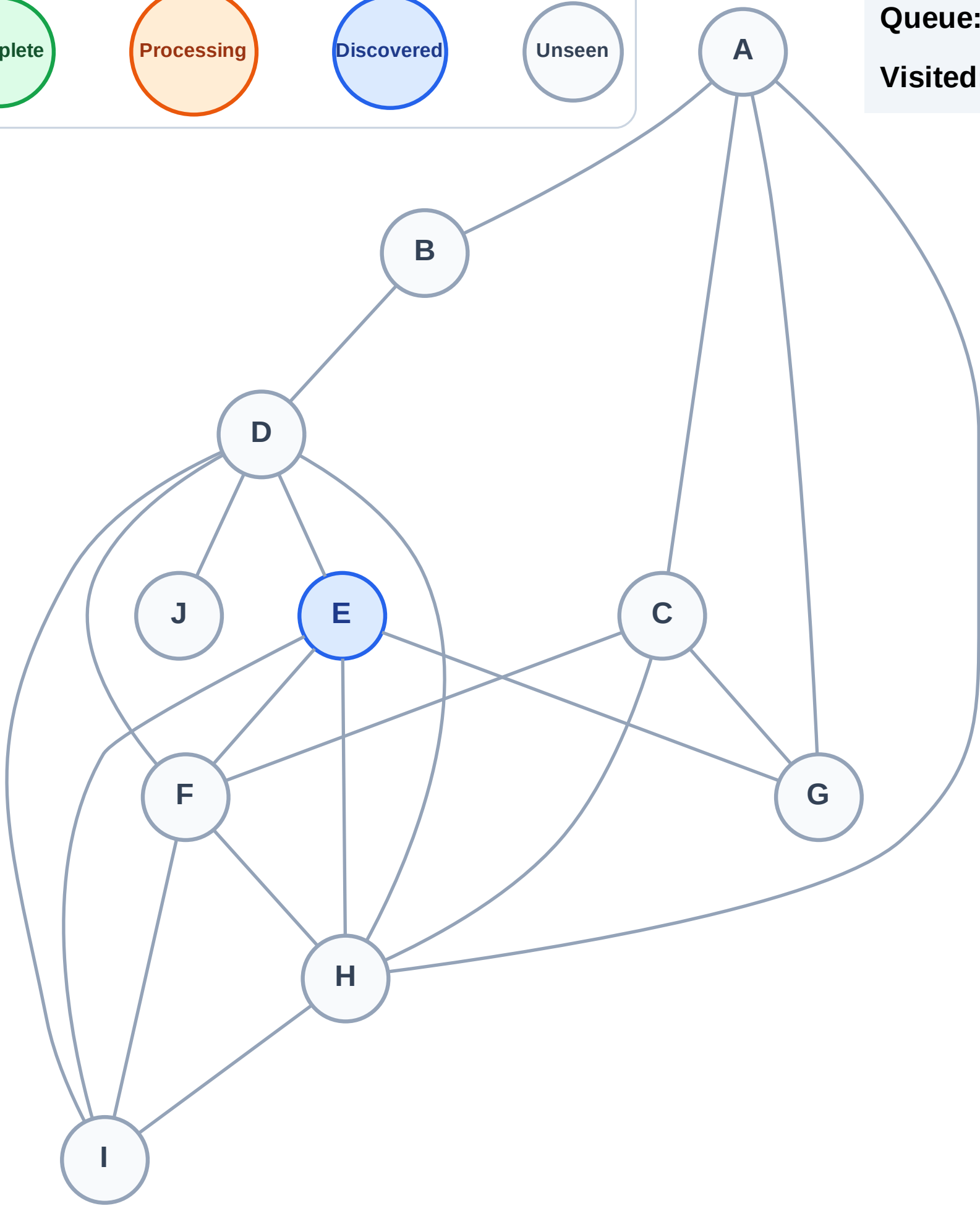
Step 1: Start at E

Legend



Queue: E

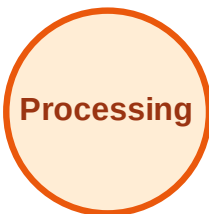
Visited: (none)



BFS Traversal

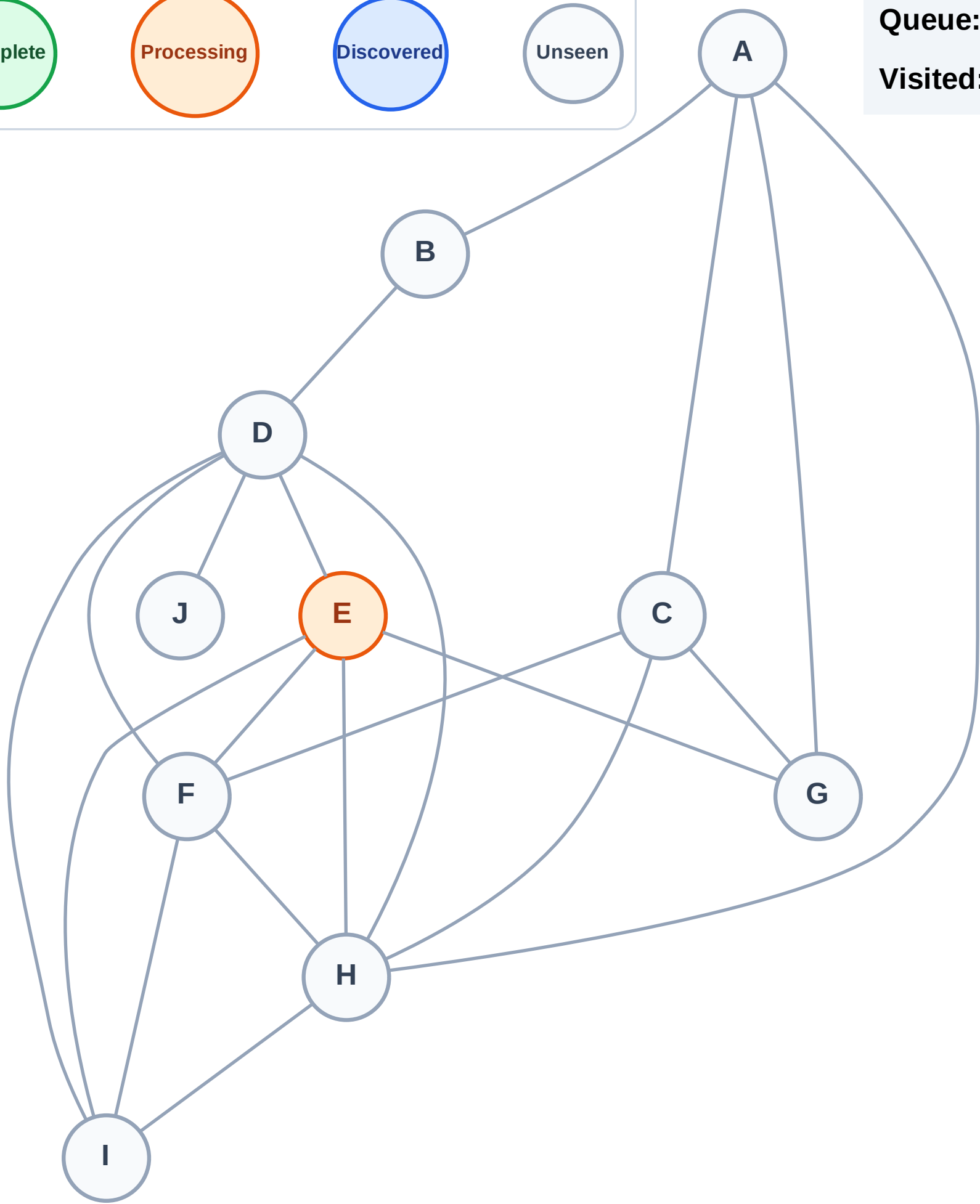
Step 2: Process node E

Legend



Queue: (empty)

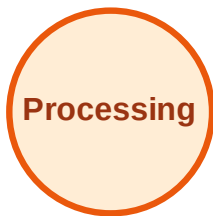
Visited: (none)



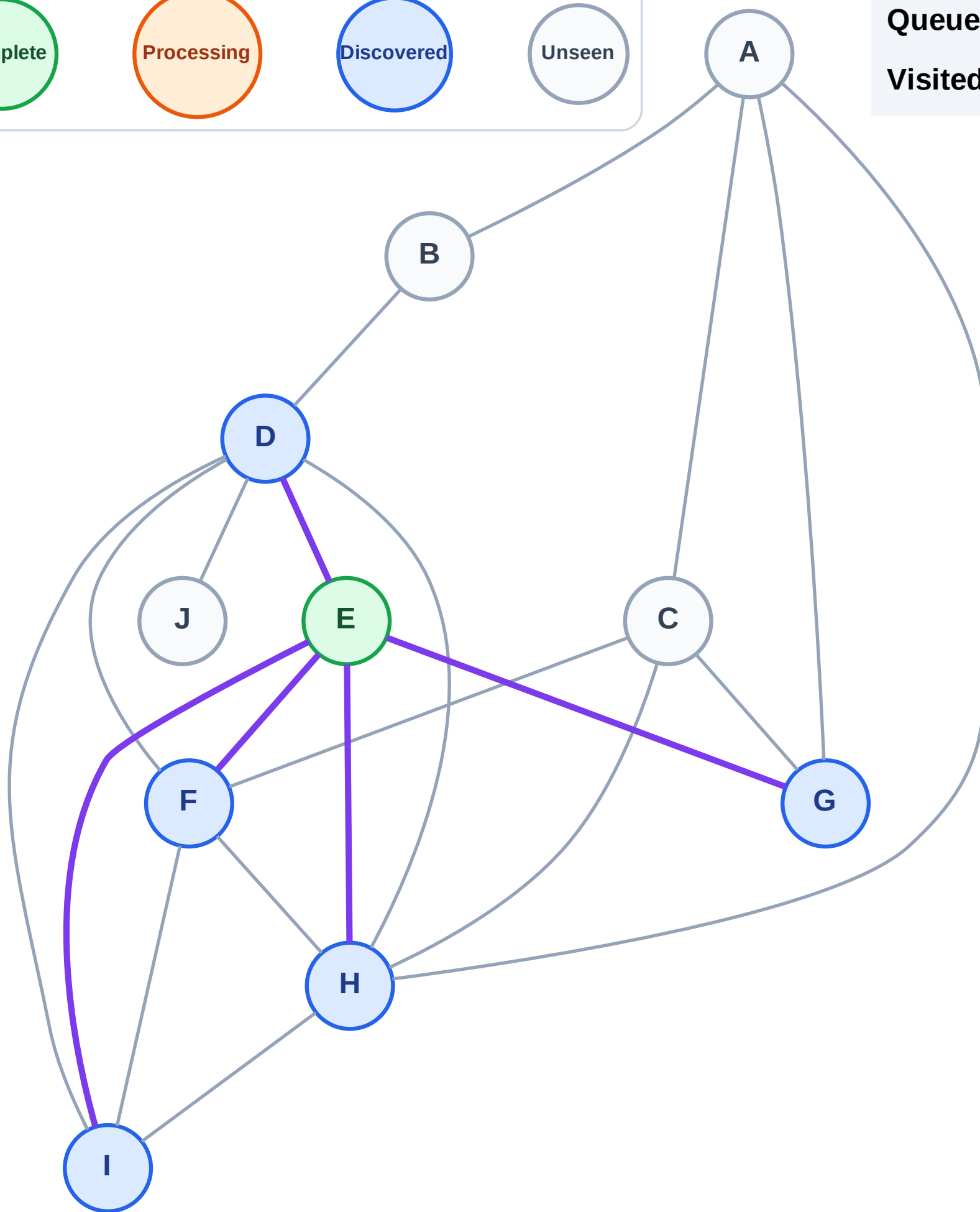
BFS Traversal

Step 3: Discover D, F, G, H, I from E

Legend



Queue: D → F → G → H → I
Visited: E



BFS Traversal

Step 4: Process node D

Legend

Complete

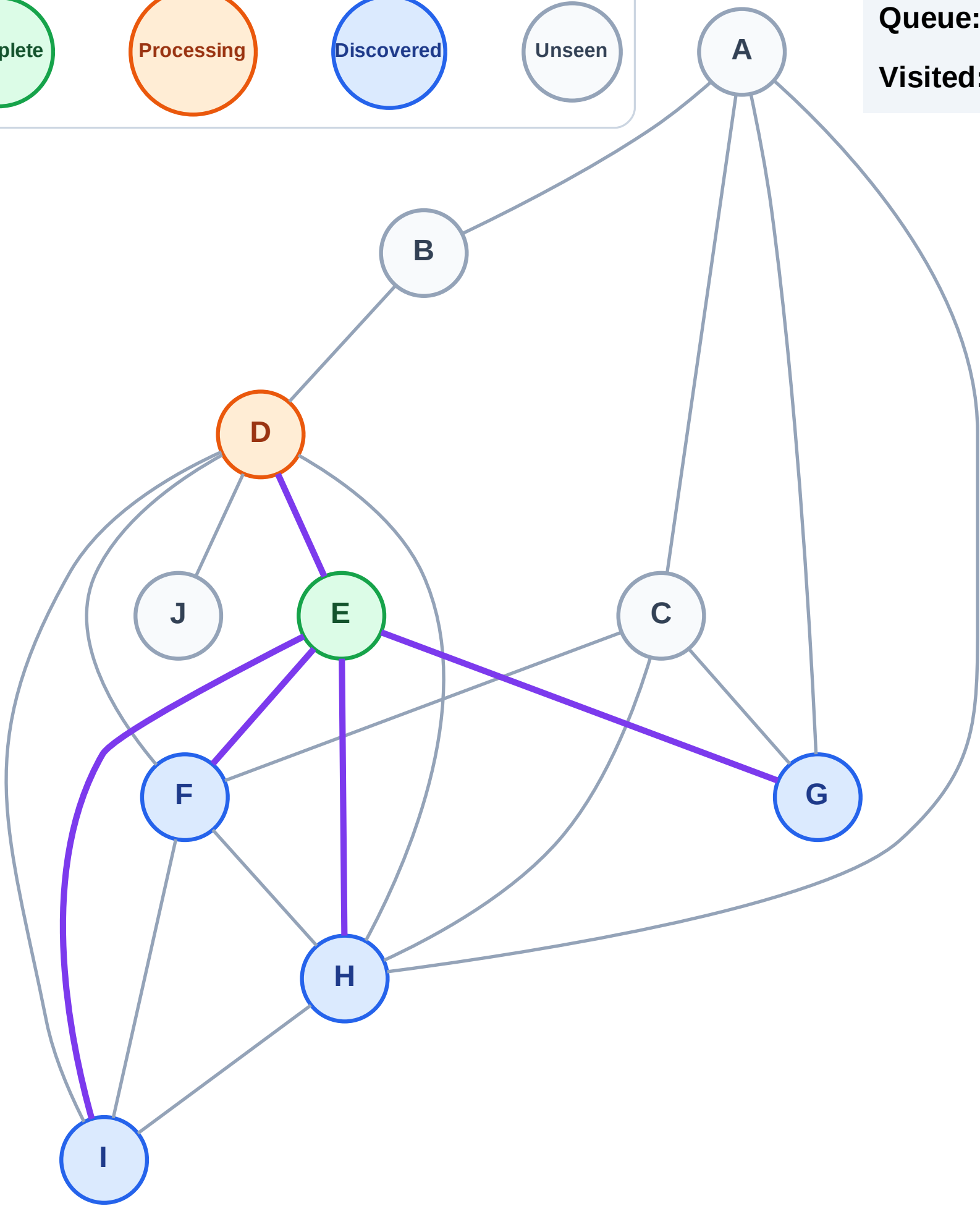
Processing

Discovered

Unseen

Queue: F → G → H → I

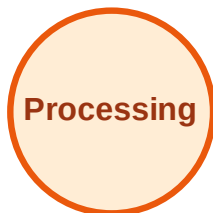
Visited: E



BFS Traversal

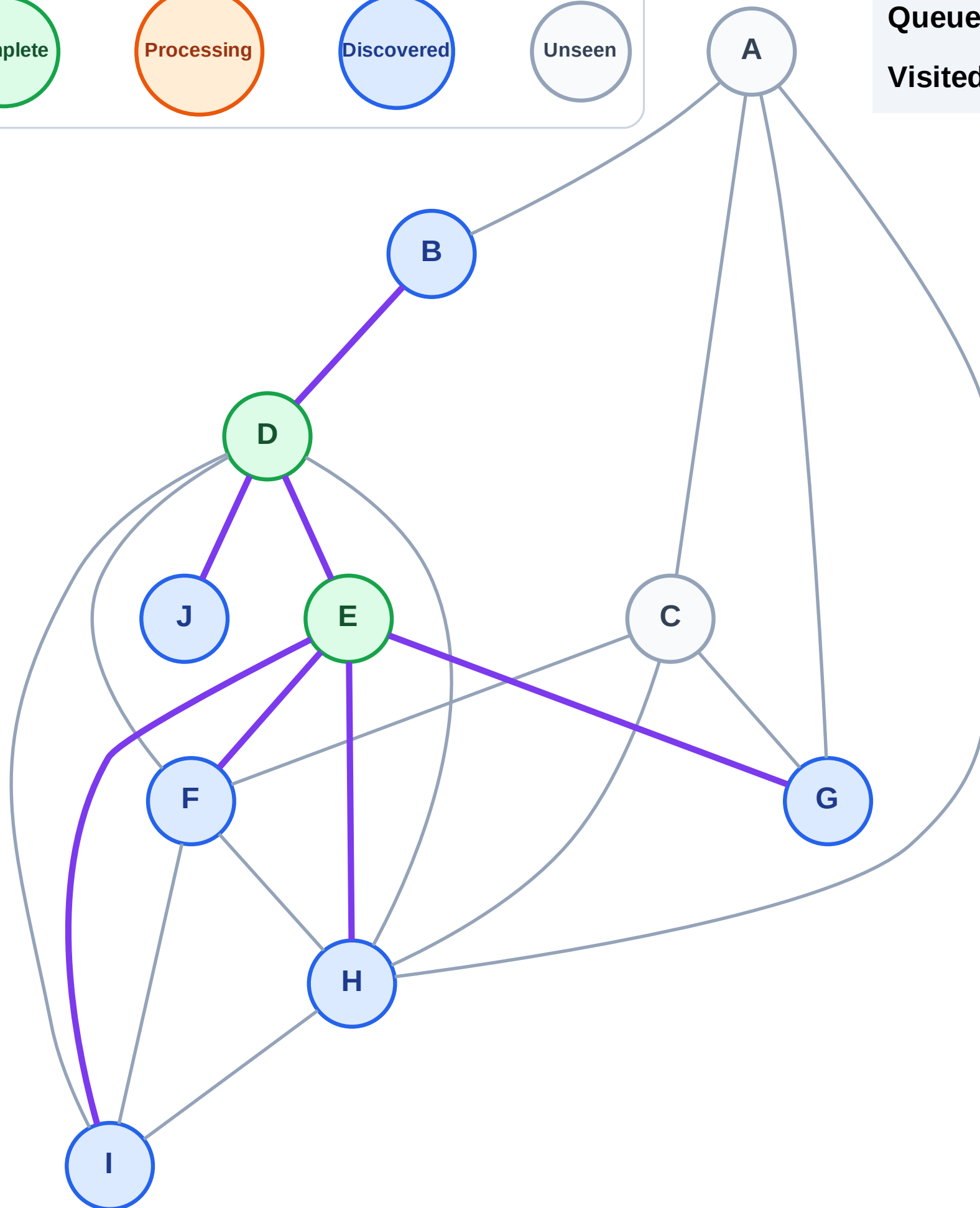
Step 5: Discover B, J from D

Legend



Queue: F → G → H → I → B → J

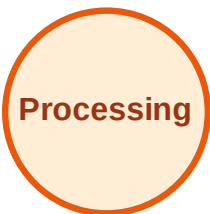
Visited: D, E



BFS Traversal

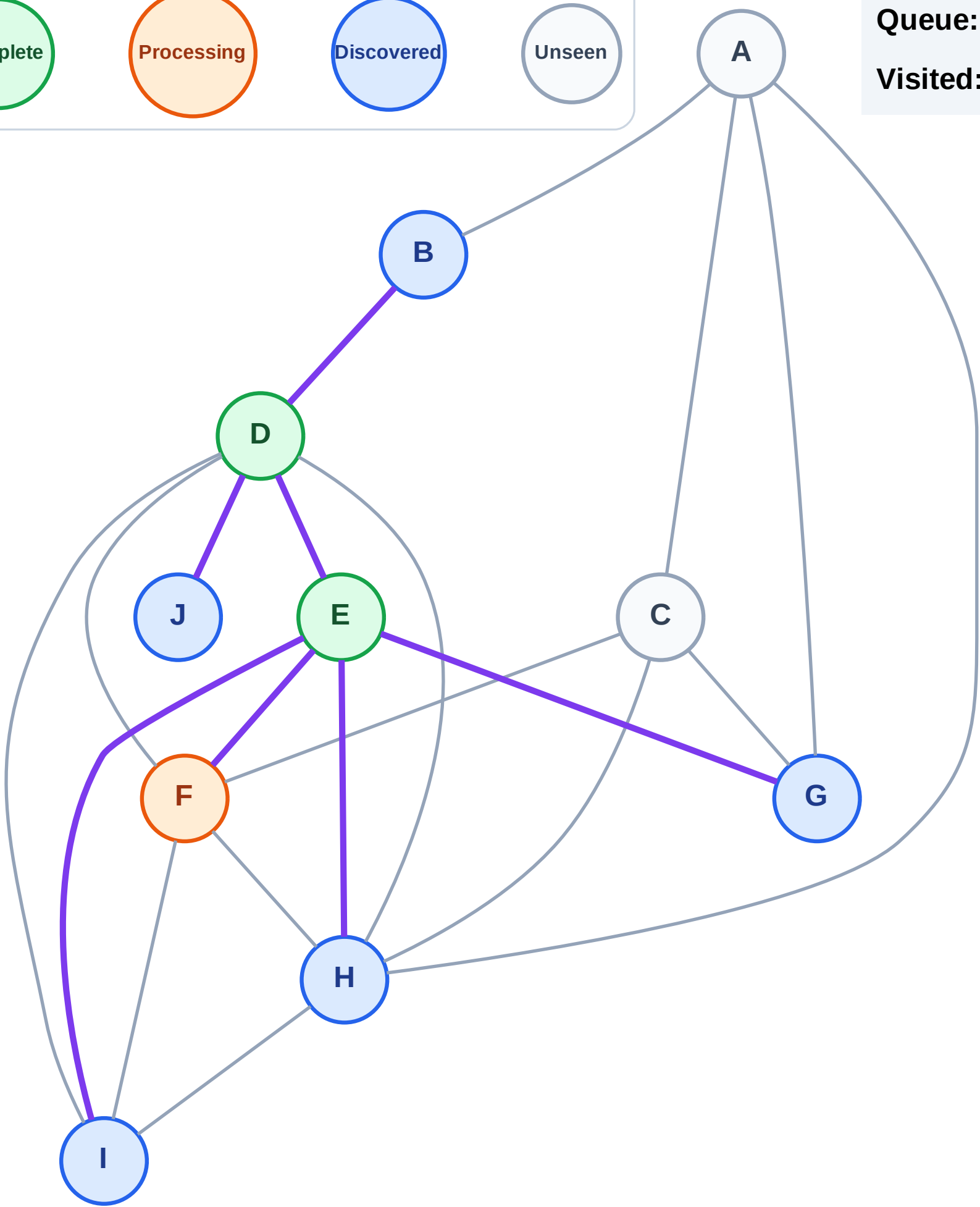
Step 6: Process node F

Legend



Queue: G → H → I → B → J

Visited: D, E



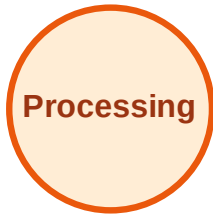
BFS Traversal

Step 7: Discover C from F

Legend



Complete



Processing



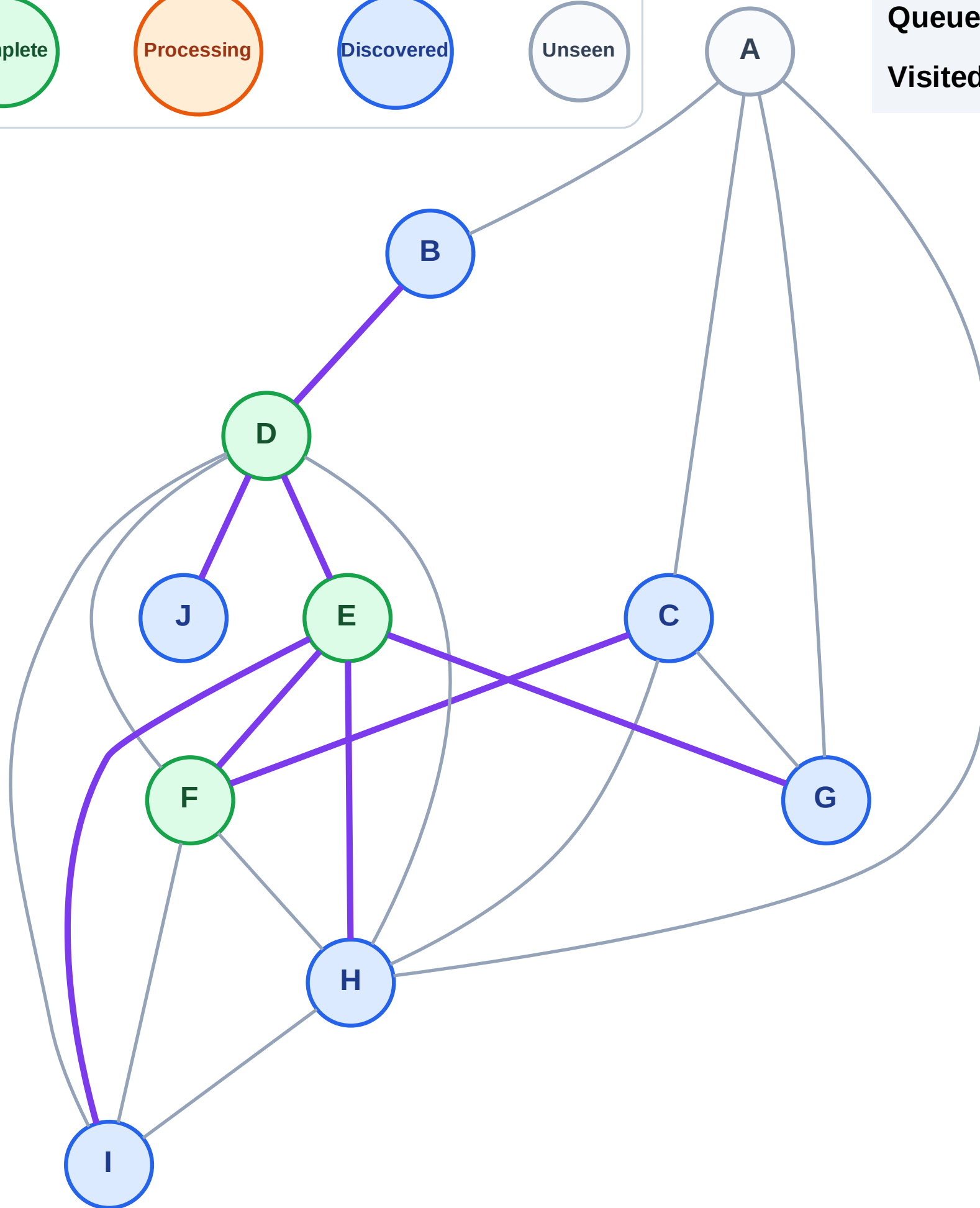
Discovered



Unseen

Queue: G → H → I → B → J → C

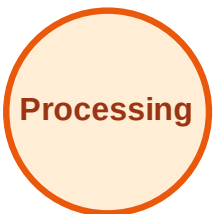
Visited: D, E, F



BFS Traversal

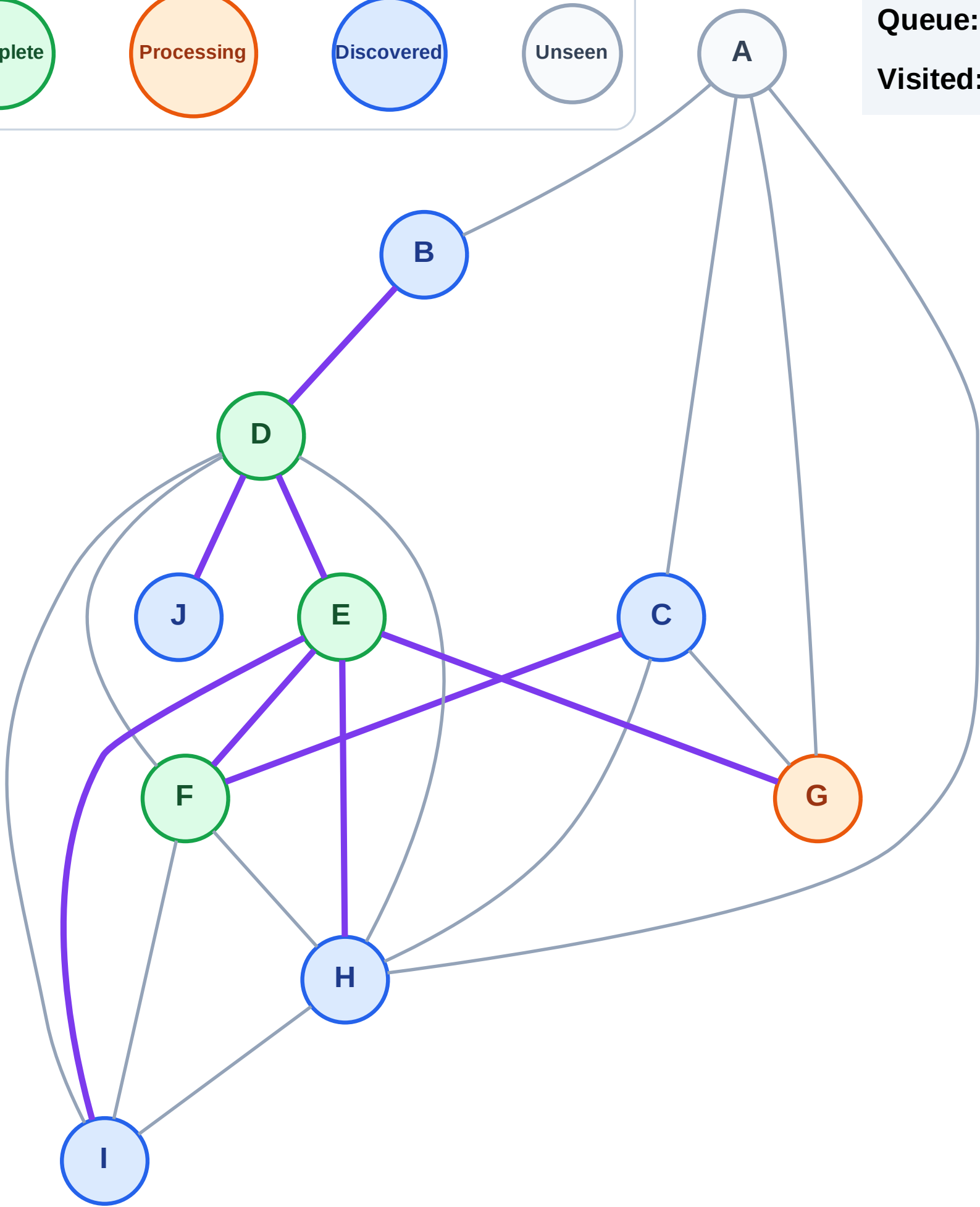
Step 8: Process node G

Legend



Queue: H → I → B → J → C

Visited: D, E, F



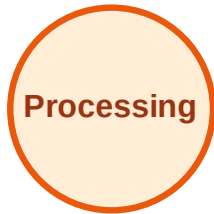
BFS Traversal

Step 9: Discover A from G

Legend



Complete



Processing

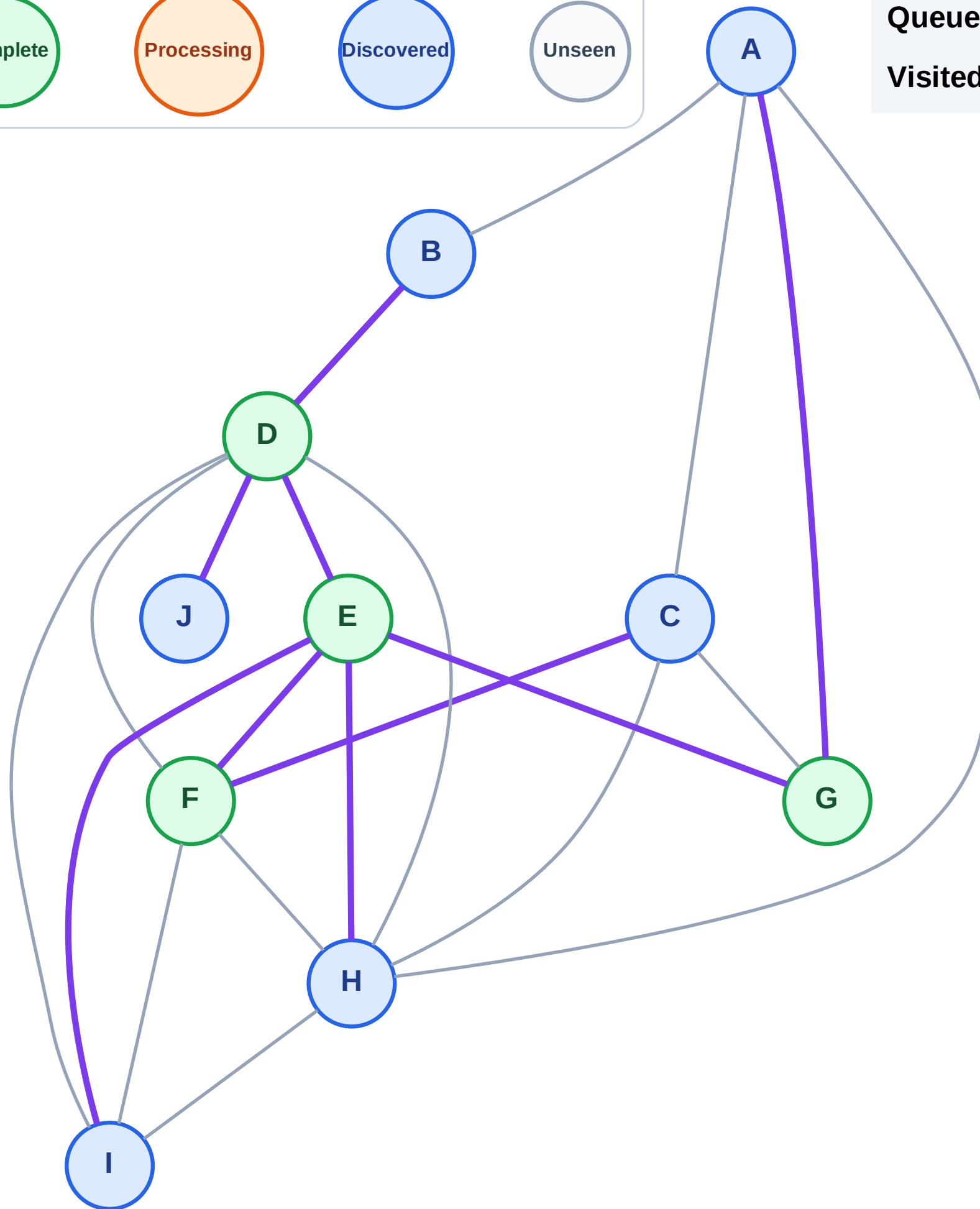


Discovered



Unseen

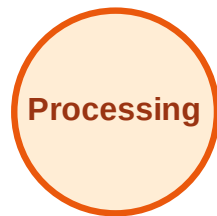
Queue: H → I → B → J → C → A
Visited: D, E, F, G



BFS Traversal

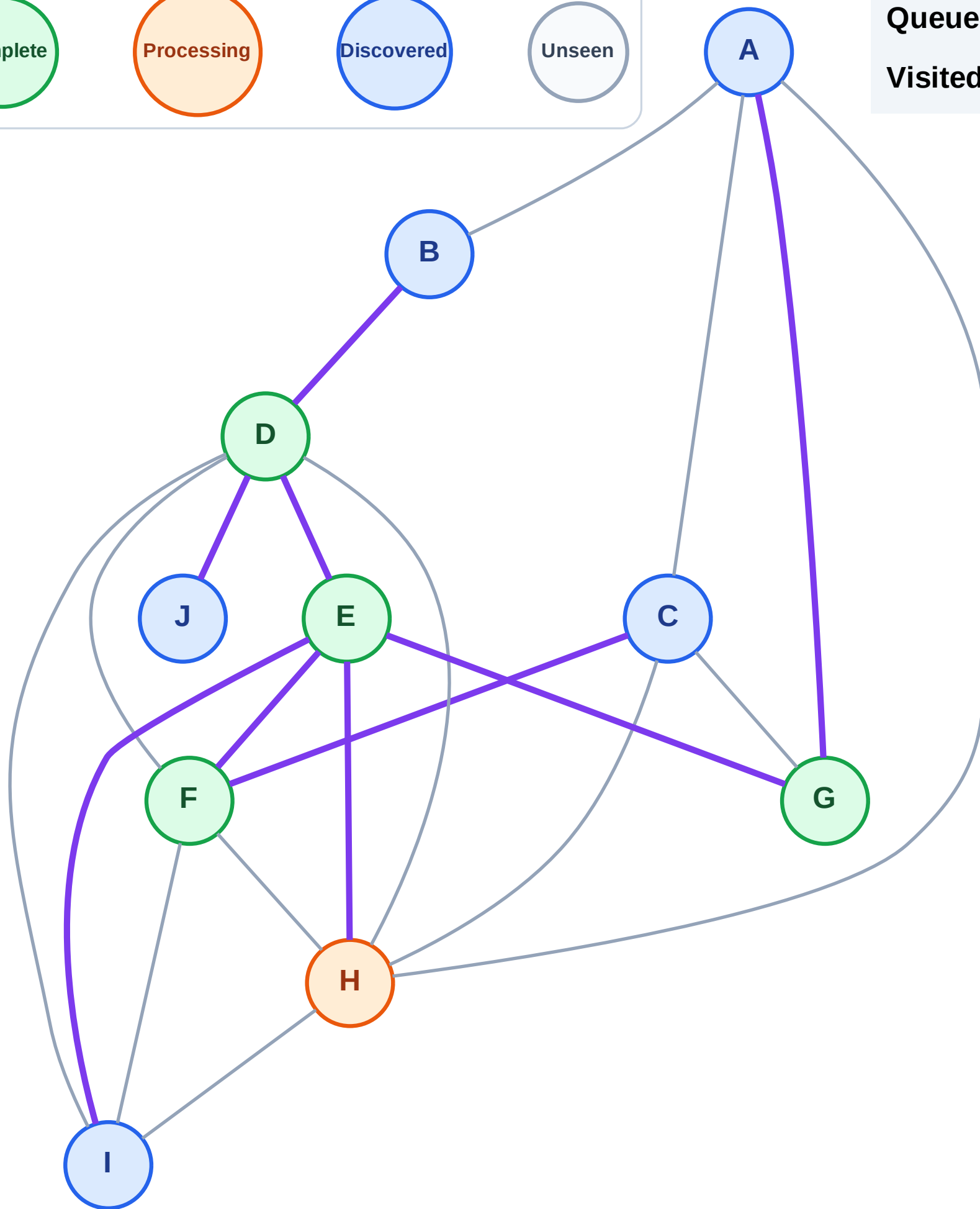
Step 10: Process node H

Legend



Queue: I → B → J → C → A

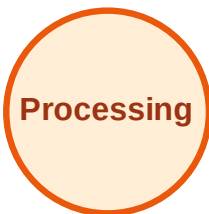
Visited: D, E, F, G



BFS Traversal

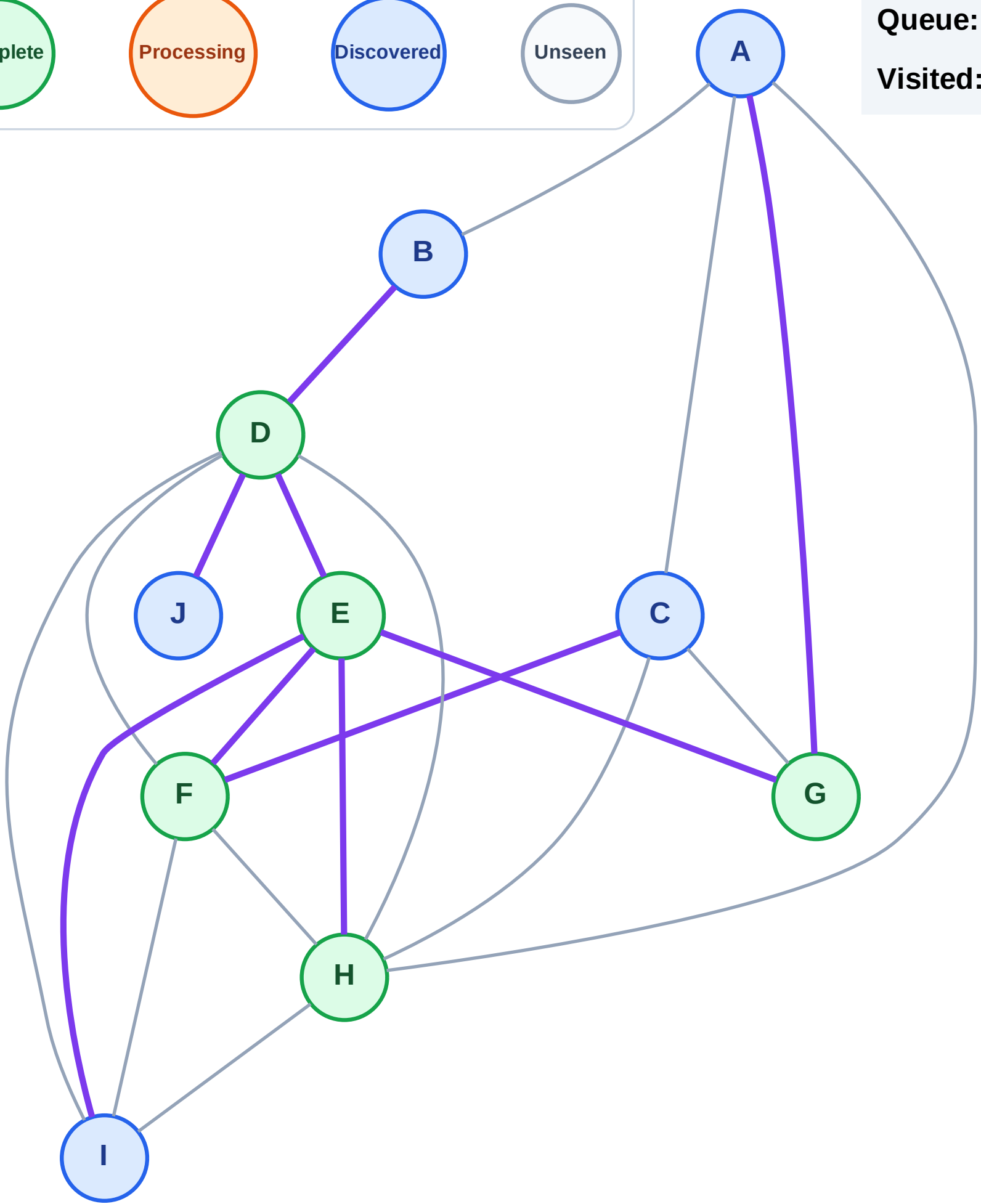
Step 11: No new neighbors from H

Legend



Queue: I → B → J → C → A

Visited: D, E, F, G, H



BFS Traversal

Step 12: Process node I

Legend

Complete

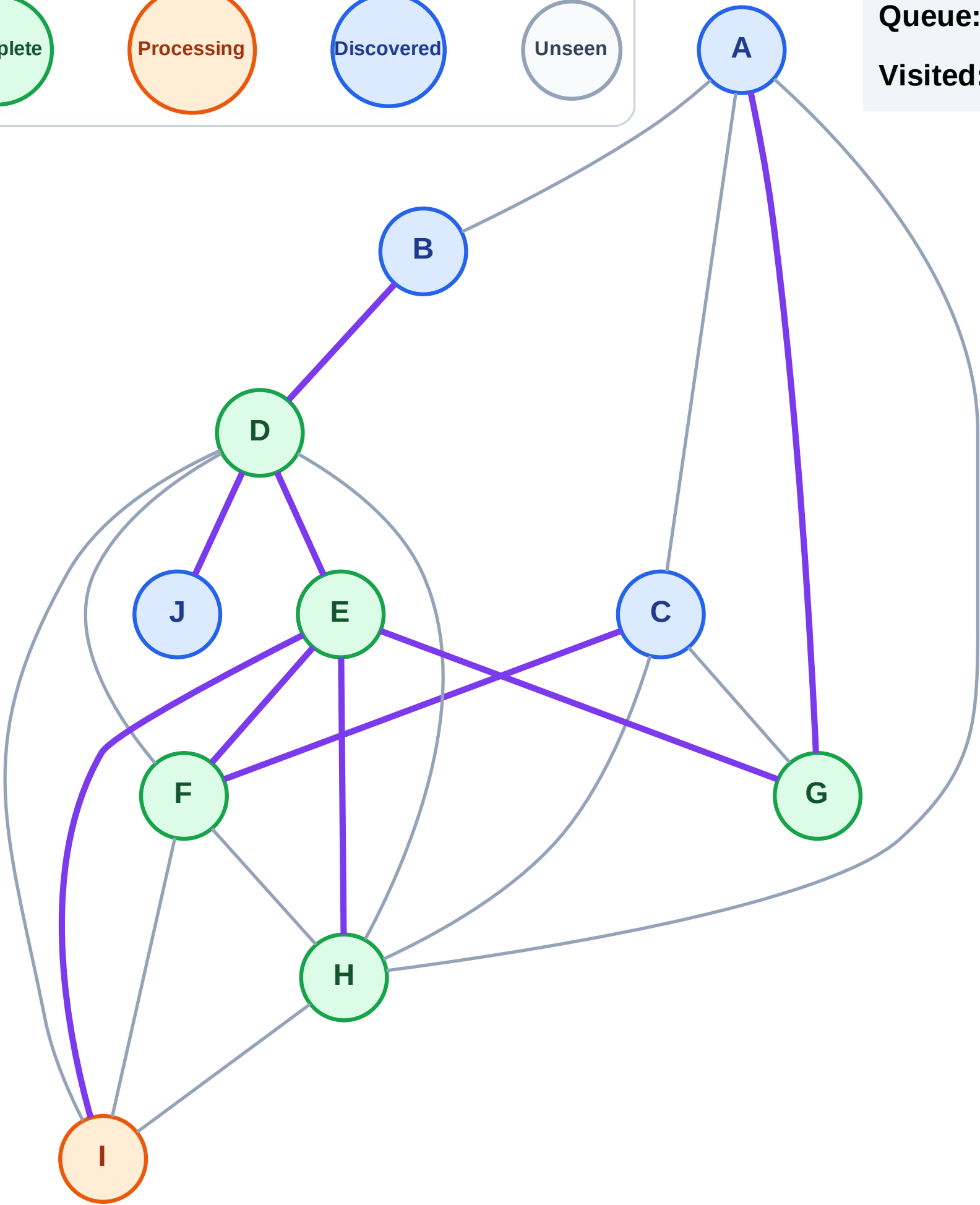
Processing

Discovered

Unseen

Queue: B → J → C → A

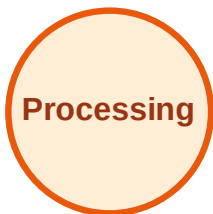
Visited: D, E, F, G, H



BFS Traversal

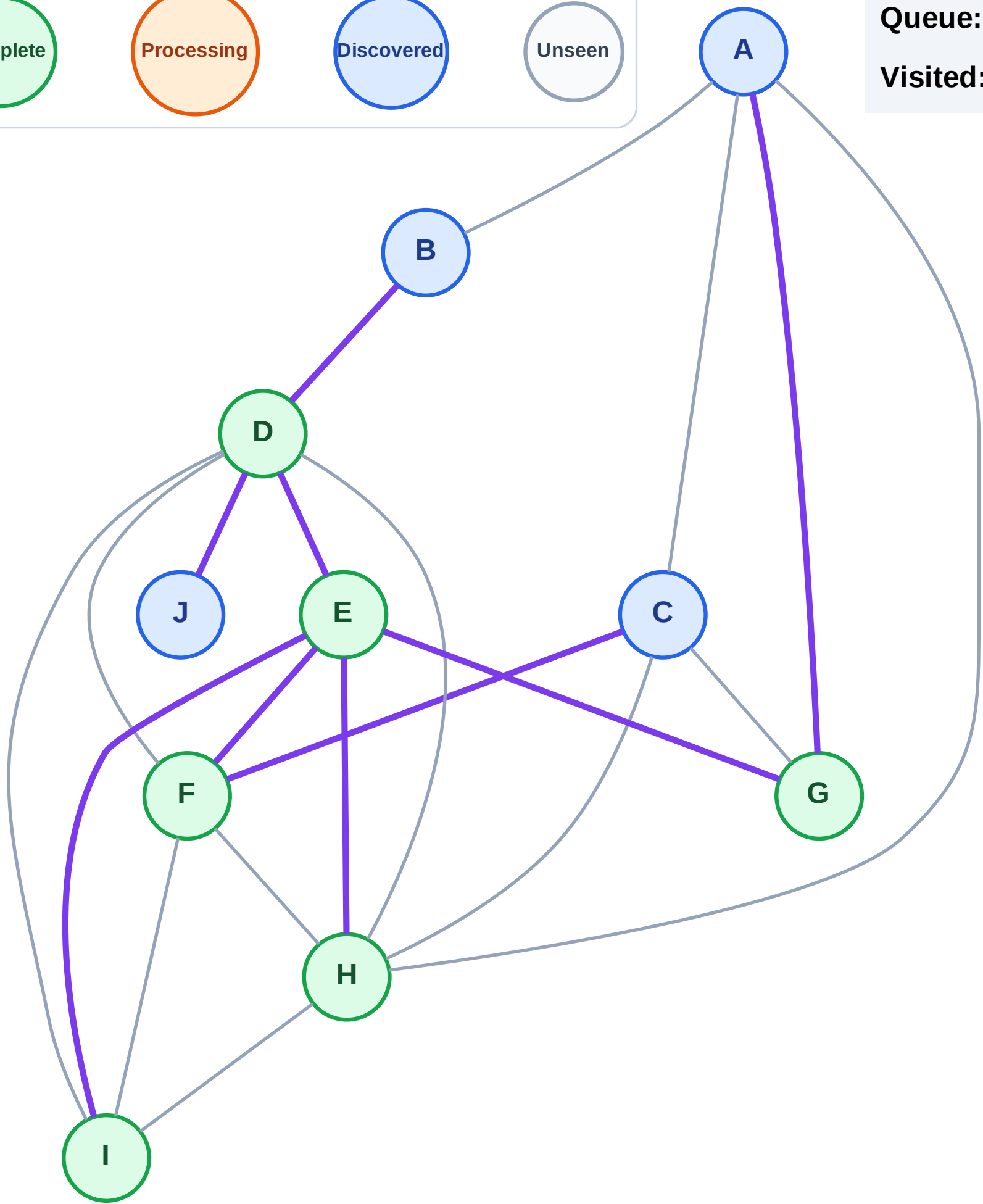
Step 13: No new neighbors from I

Legend



Queue: B → J → C → A

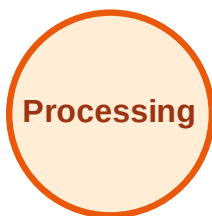
Visited: D, E, F, G, H, I



BFS Traversal

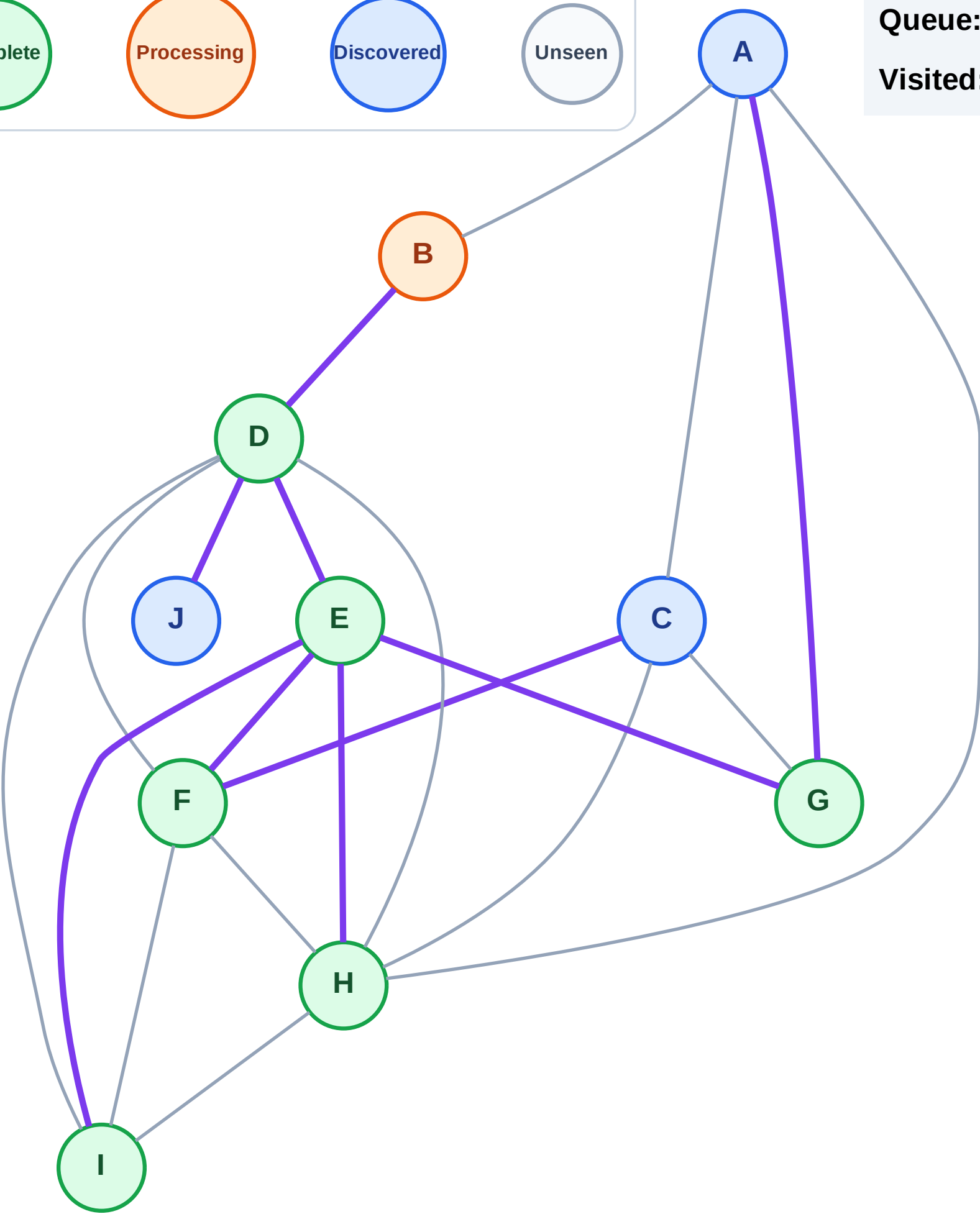
Step 14: Process node B

Legend



Queue: J → C → A

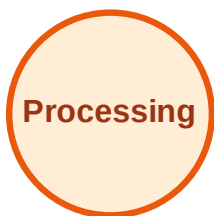
Visited: D, E, F, G, H, I



BFS Traversal

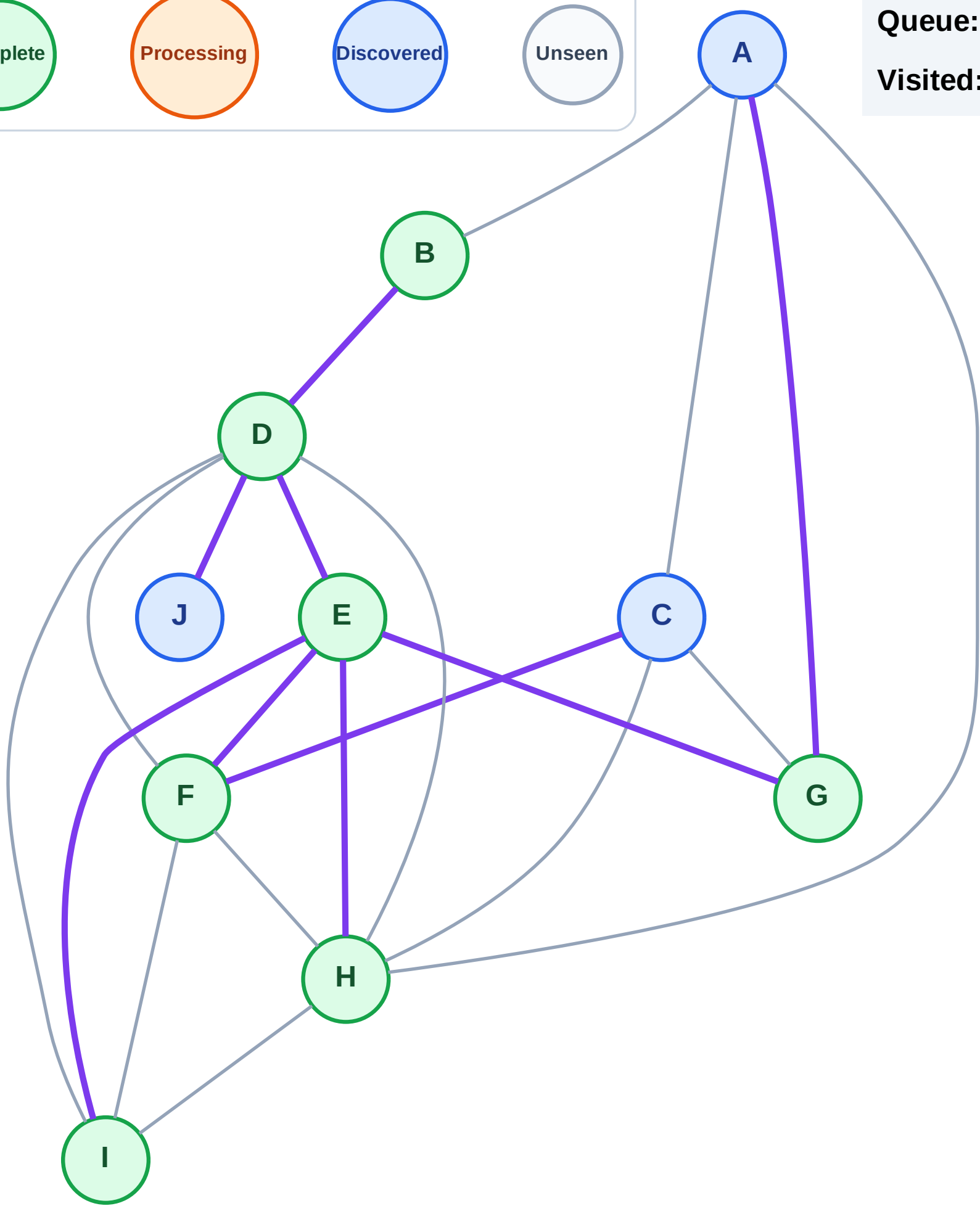
Step 15: No new neighbors from B

Legend



Queue: J → C → A

Visited: B, D, E, F, G, H, I



BFS Traversal

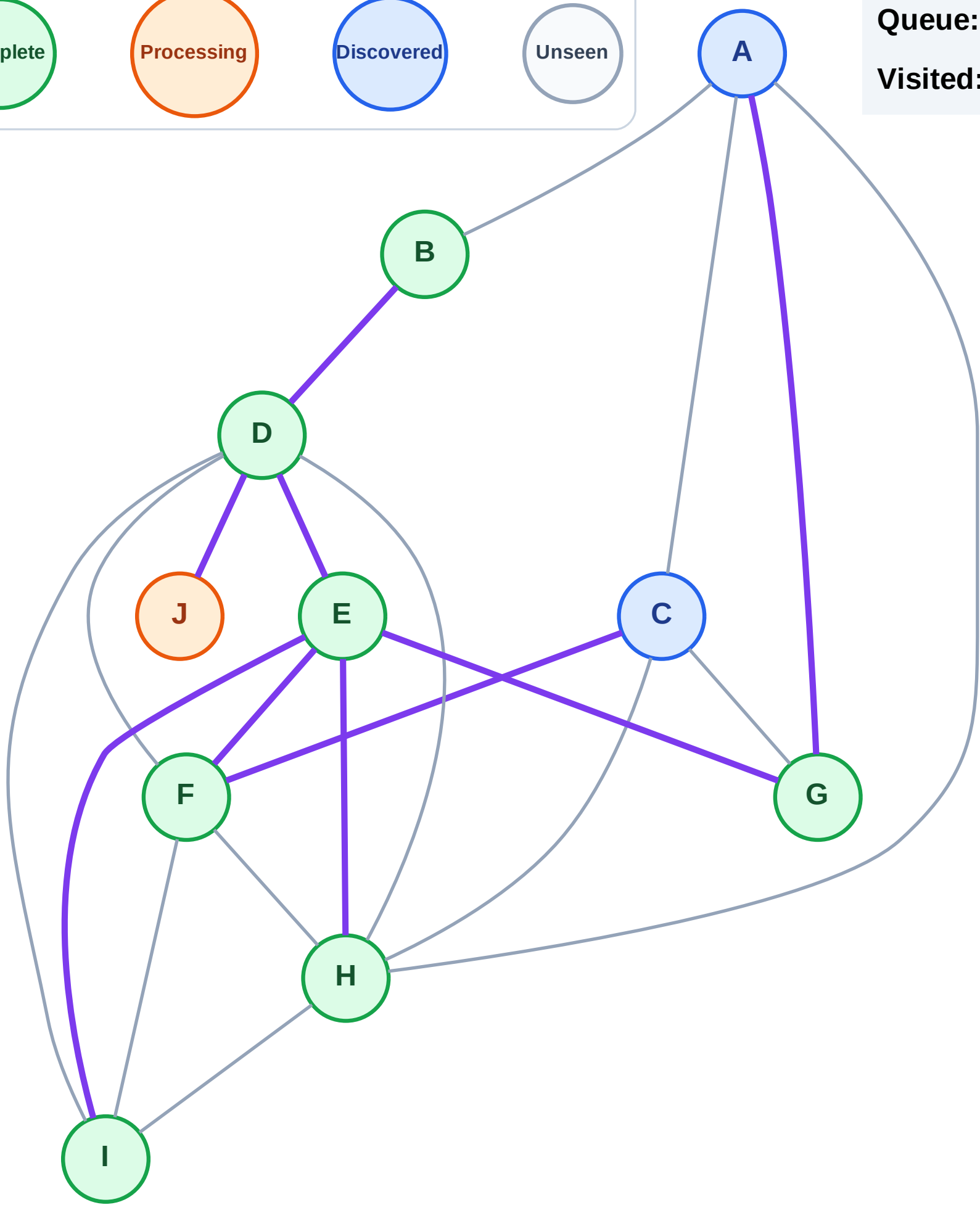
Step 16: Process node J

Legend



Queue: C → A

Visited: B, D, E, F, G, H, I



BFS Traversal

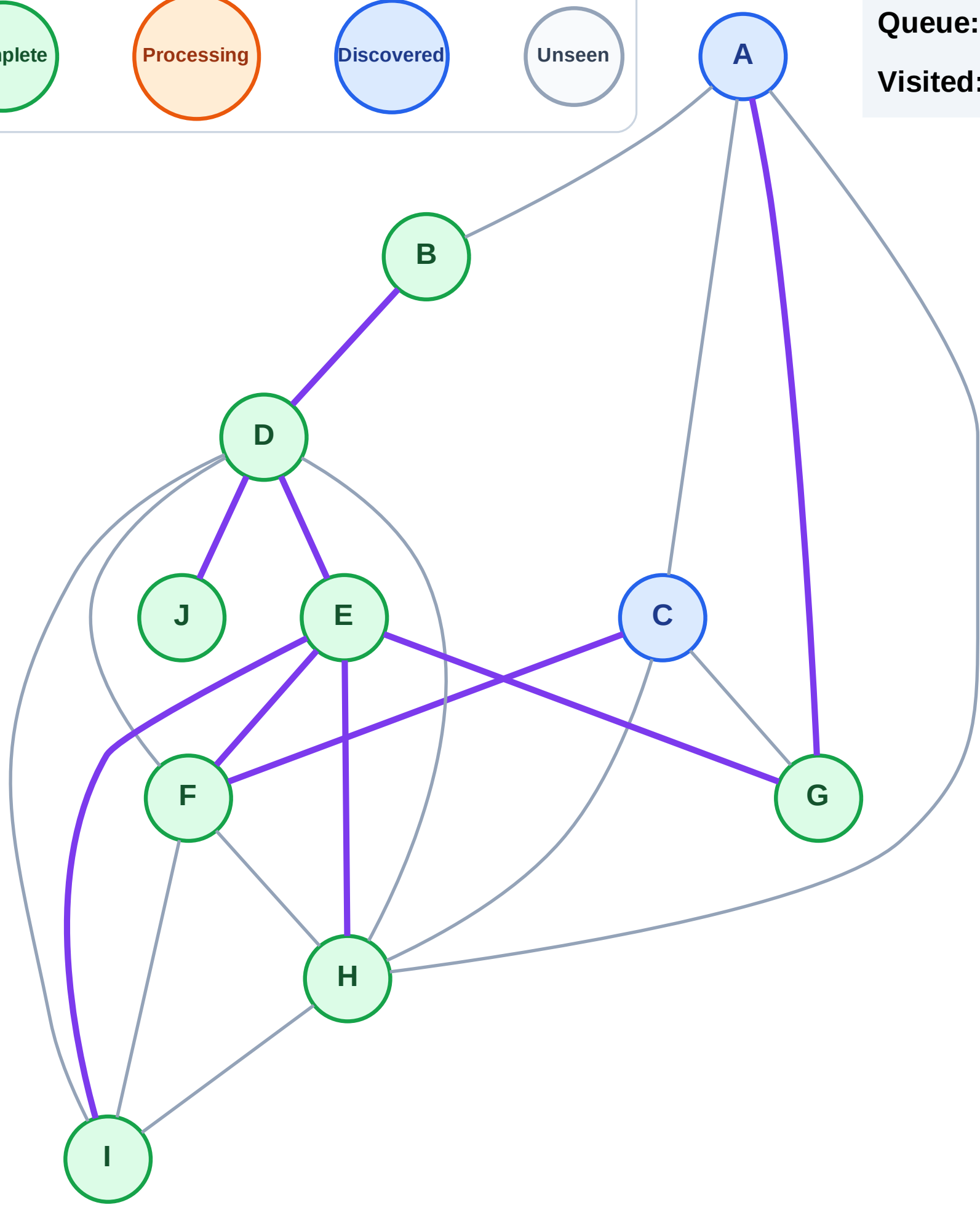
Step 17: No new neighbors from J

Legend



Queue: C → A

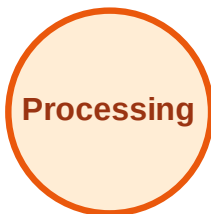
Visited: B, D, E, F, G, H, I, J



BFS Traversal

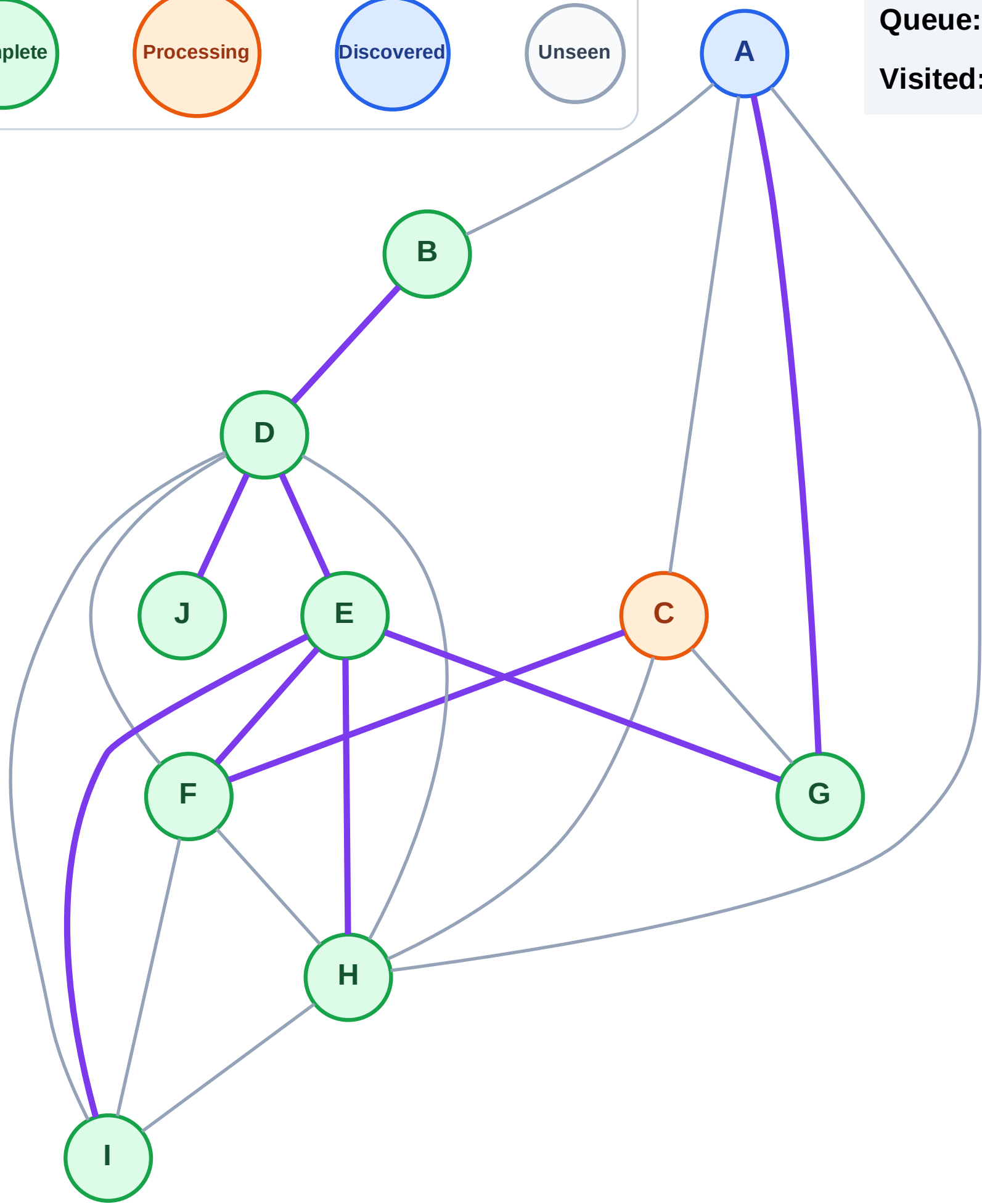
Step 18: Process node C

Legend



Queue: A

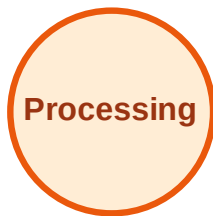
Visited: B, D, E, F, G, H, I, J



BFS Traversal

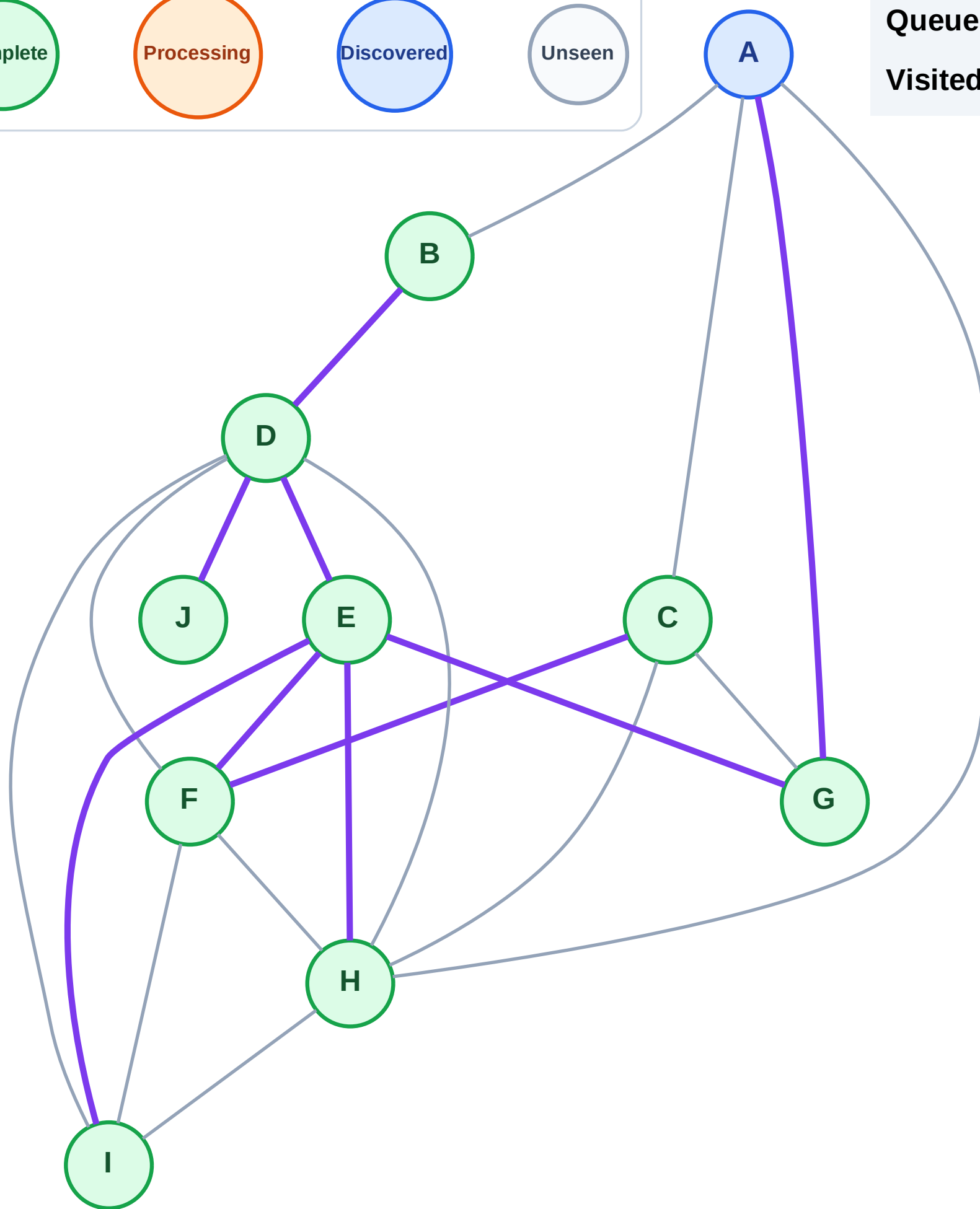
Step 19: No new neighbors from C

Legend



Queue: A

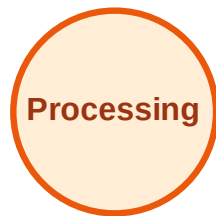
Visited: B, C, D, E, F, G, H, I, J



BFS Traversal

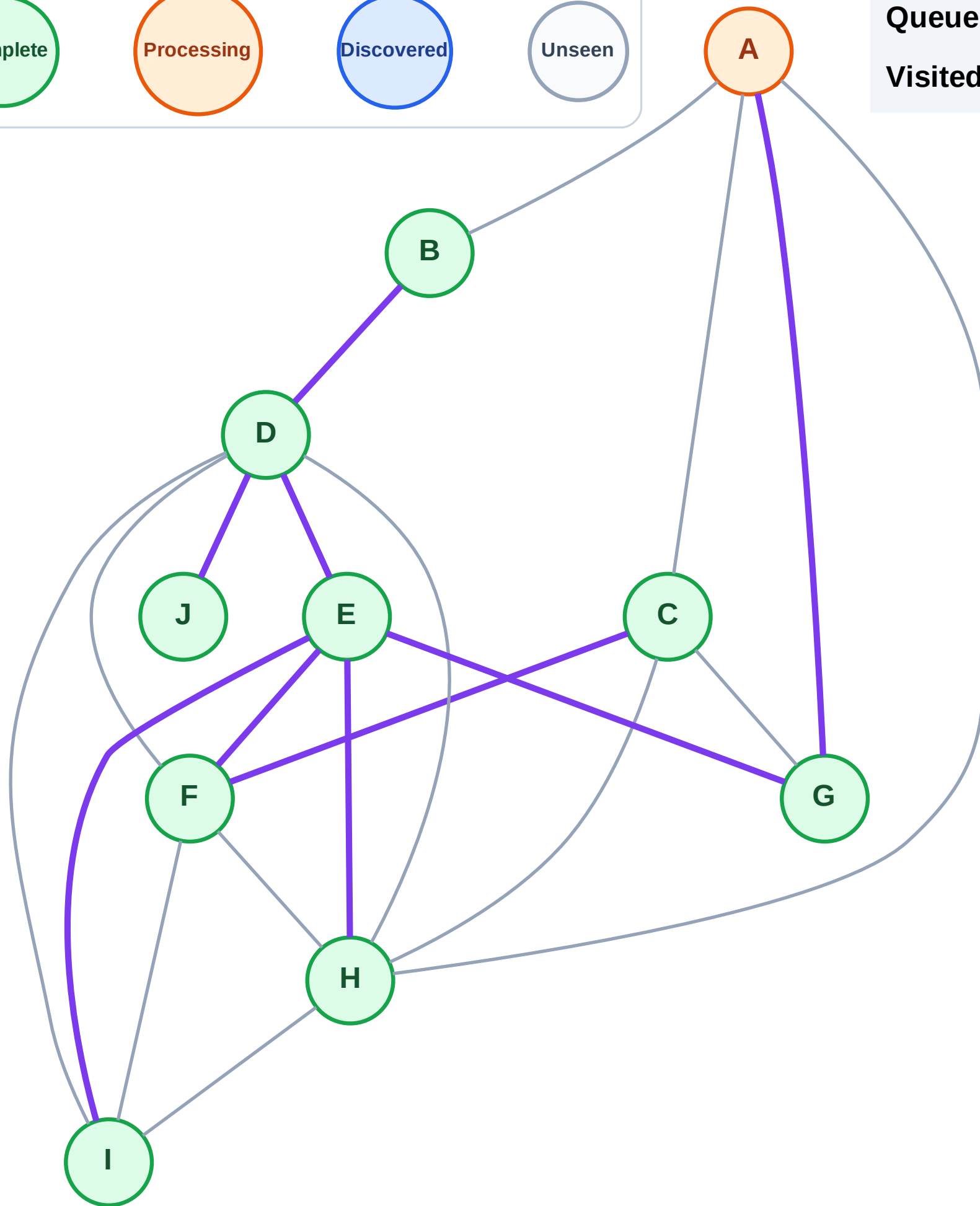
Step 20: Process node A

Legend



Queue: (empty)

Visited: B, C, D, E, F, G, H, I, J



BFS Traversal

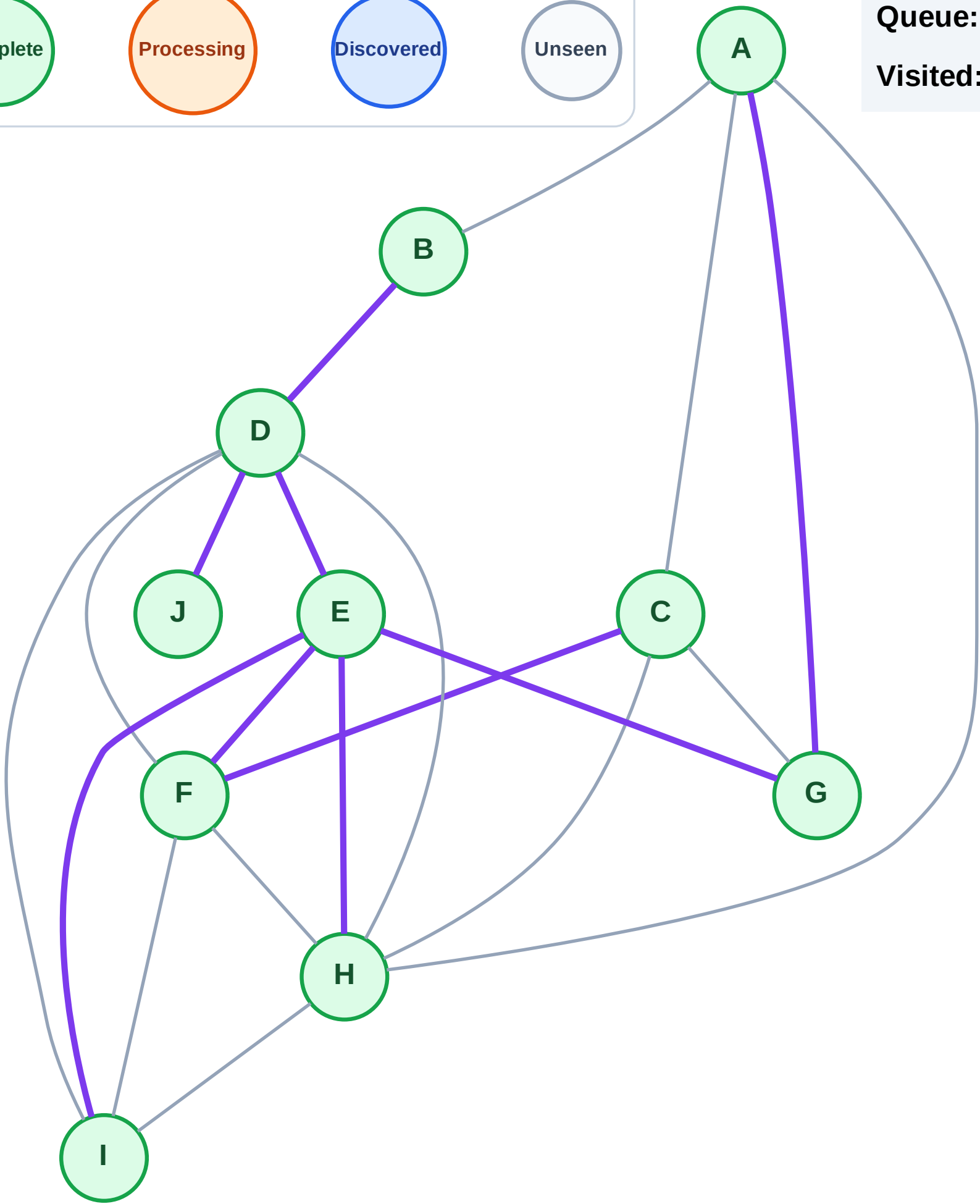
Step 21: No new neighbors from A

Legend



Queue: (empty)

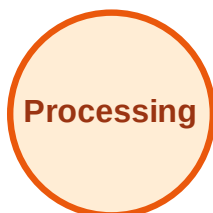
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

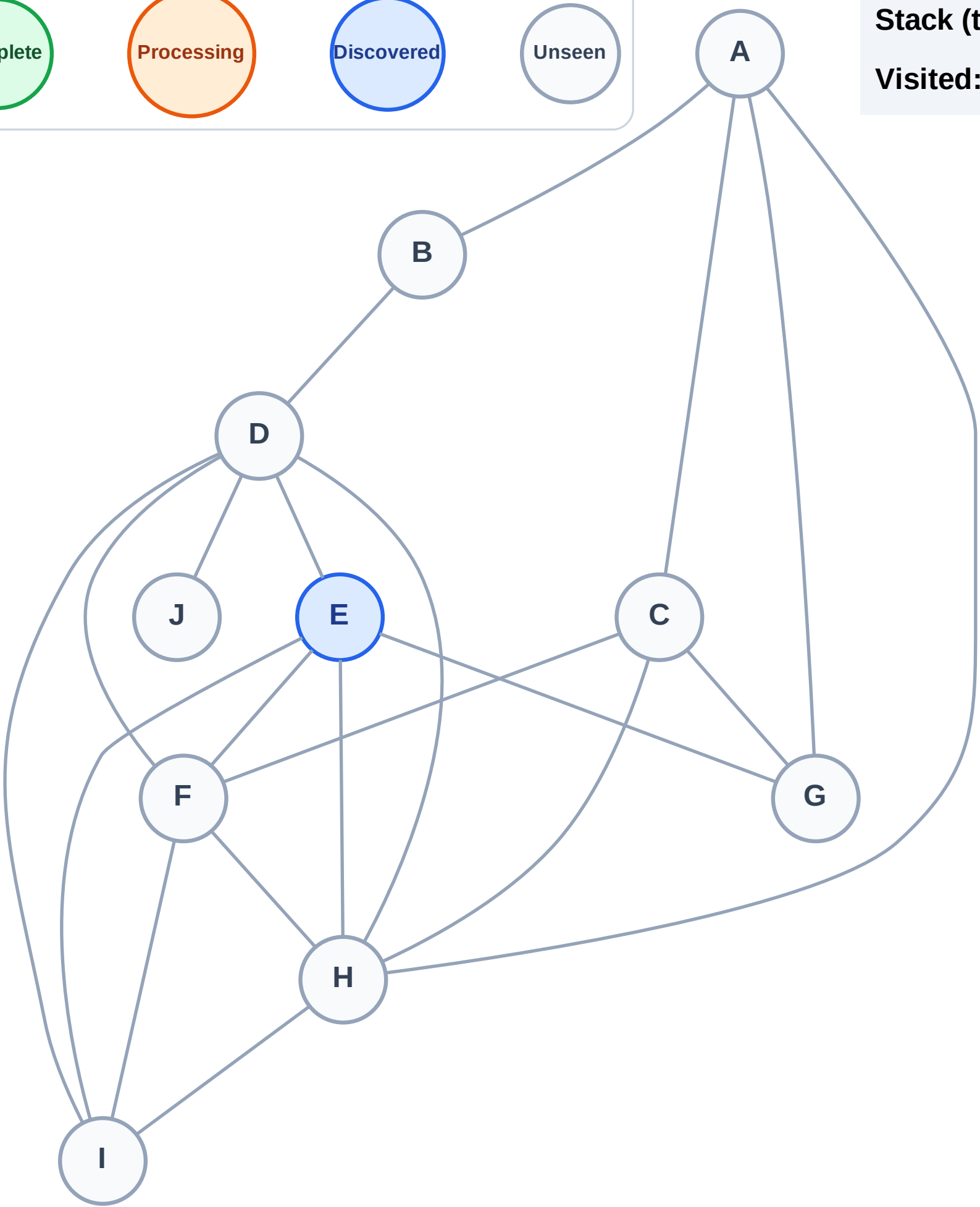
Step 1: Start at E

Legend



Stack (top at right): E

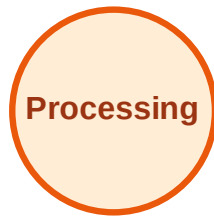
Visited: (none)



DFS Traversal

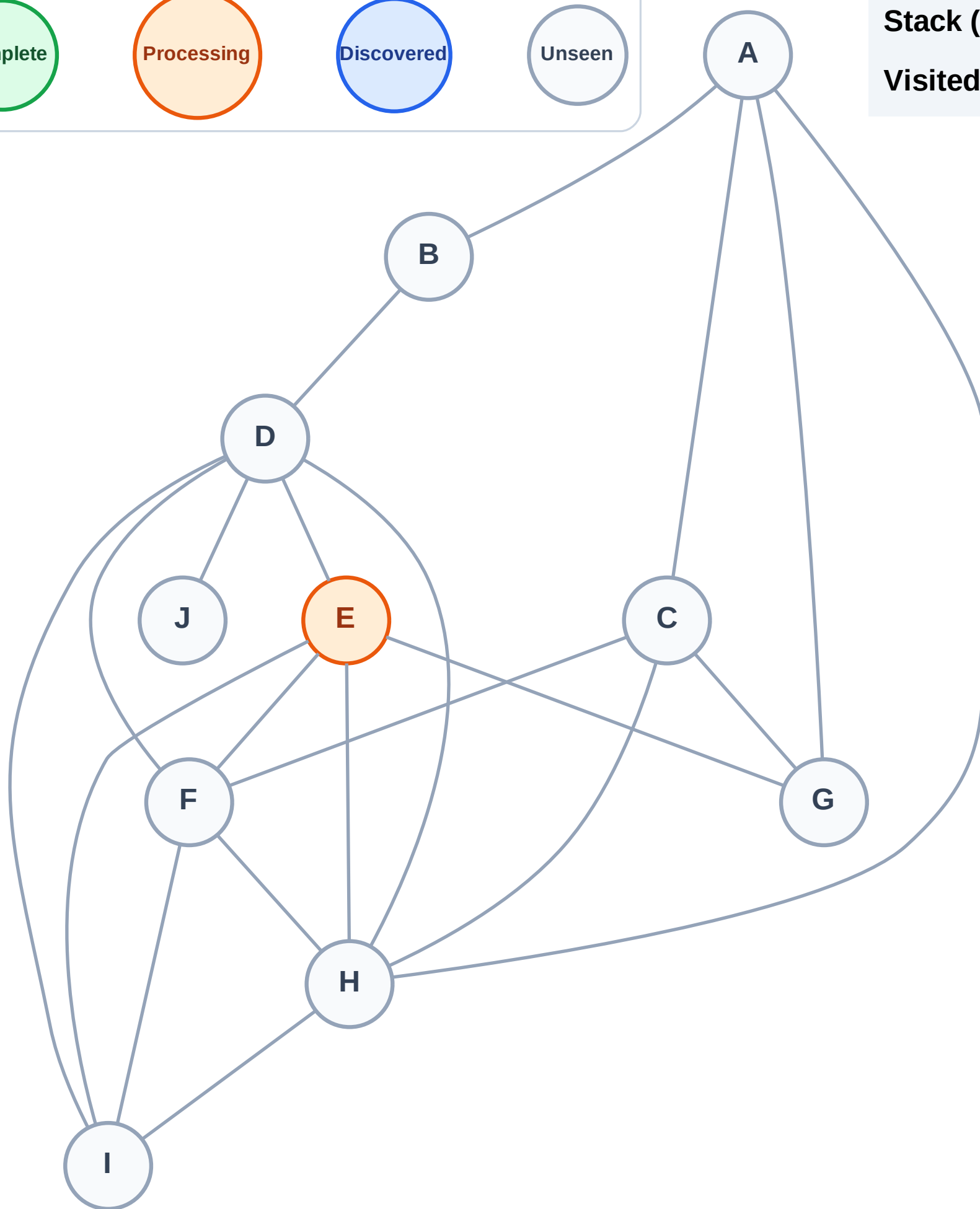
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



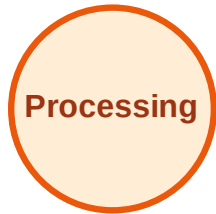
DFS Traversal

Step 3: Discover I, H, G, F, D from E

Legend



Complete



Processing



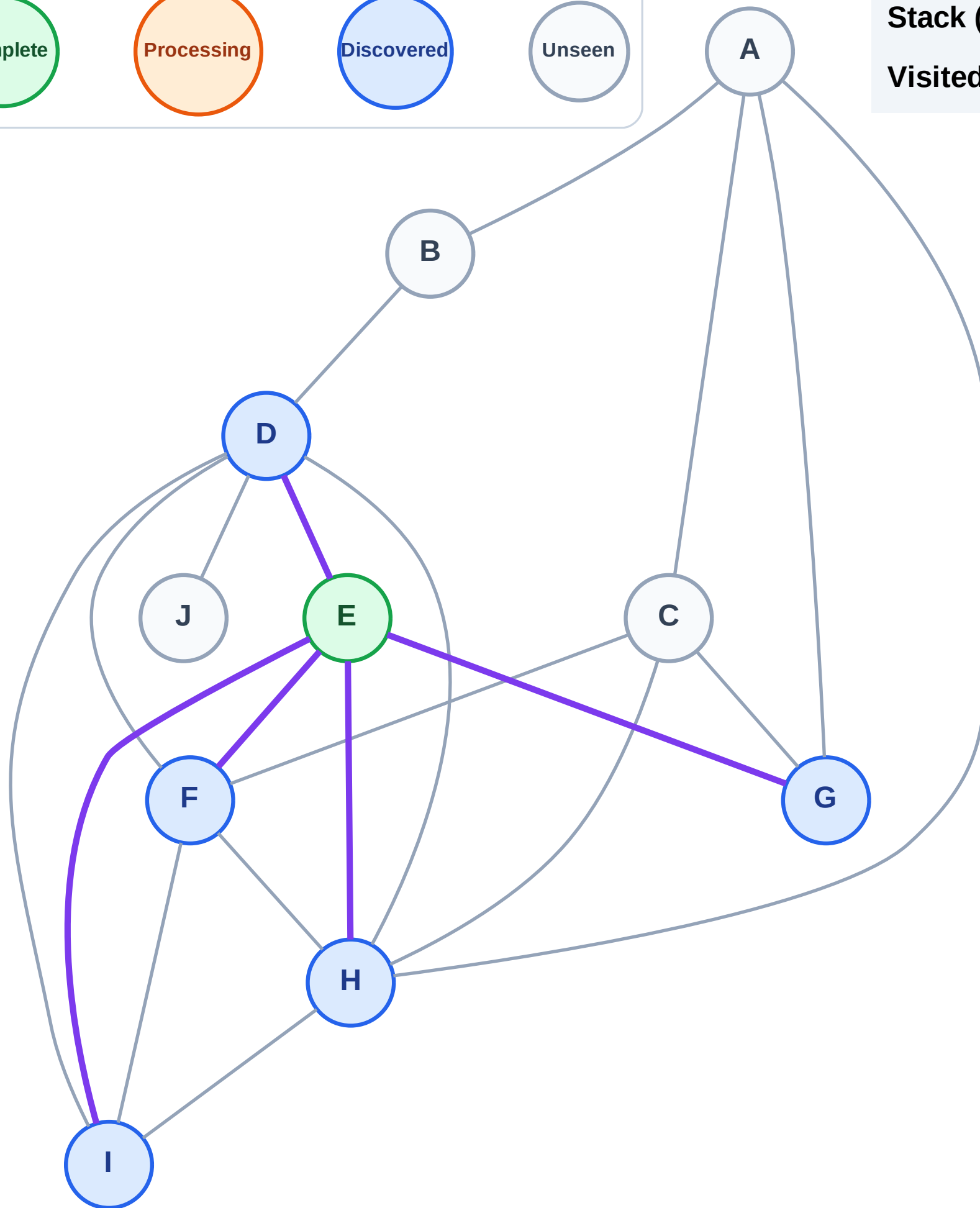
Discovered



Unseen

Stack (top at right): I | H | G | F | D

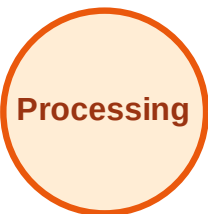
Visited: E



DFS Traversal

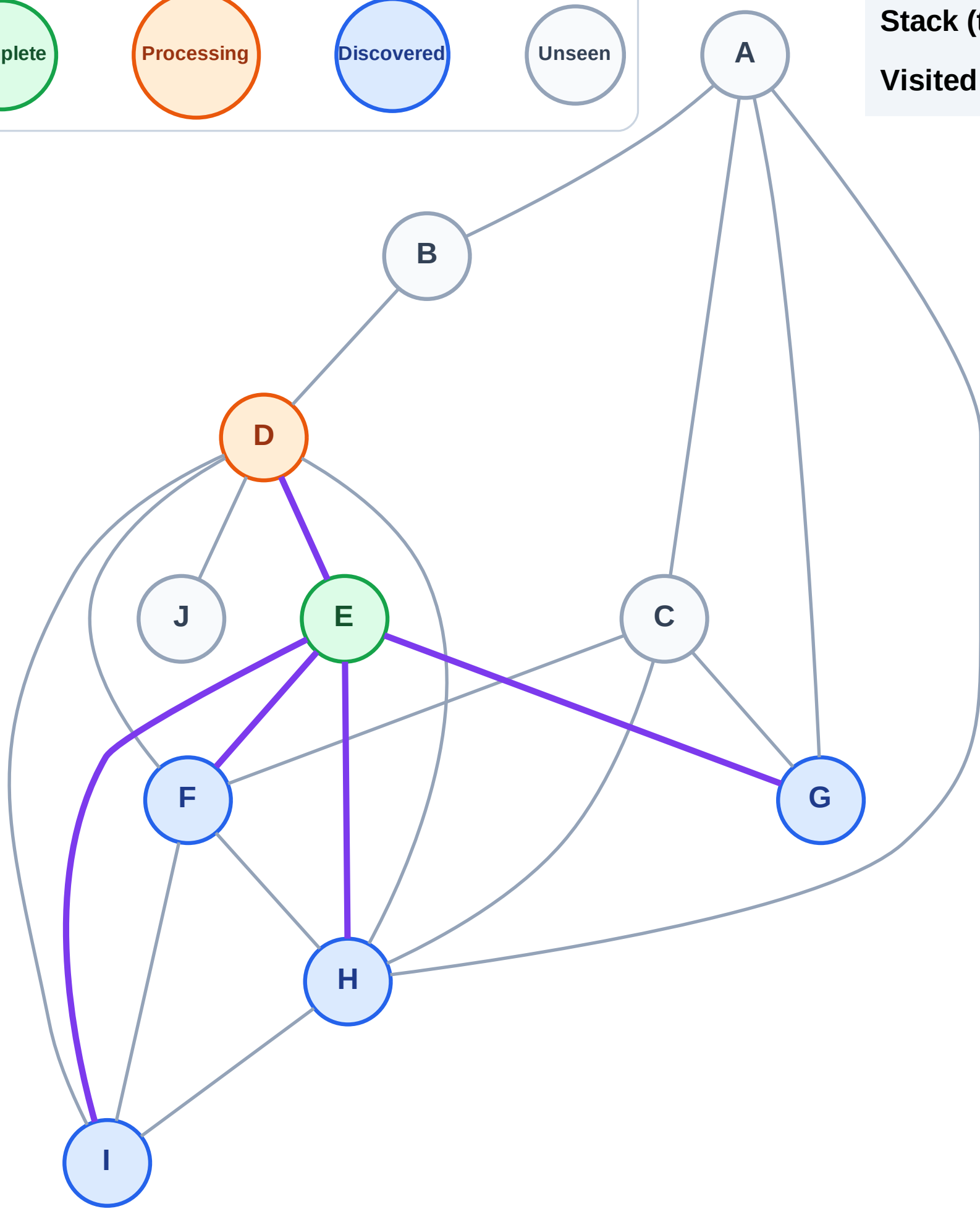
Step 4: Process node D

Legend



Stack (top at right): I | H | G | F

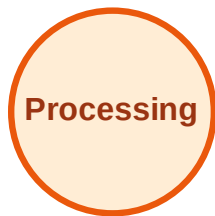
Visited: E



DFS Traversal

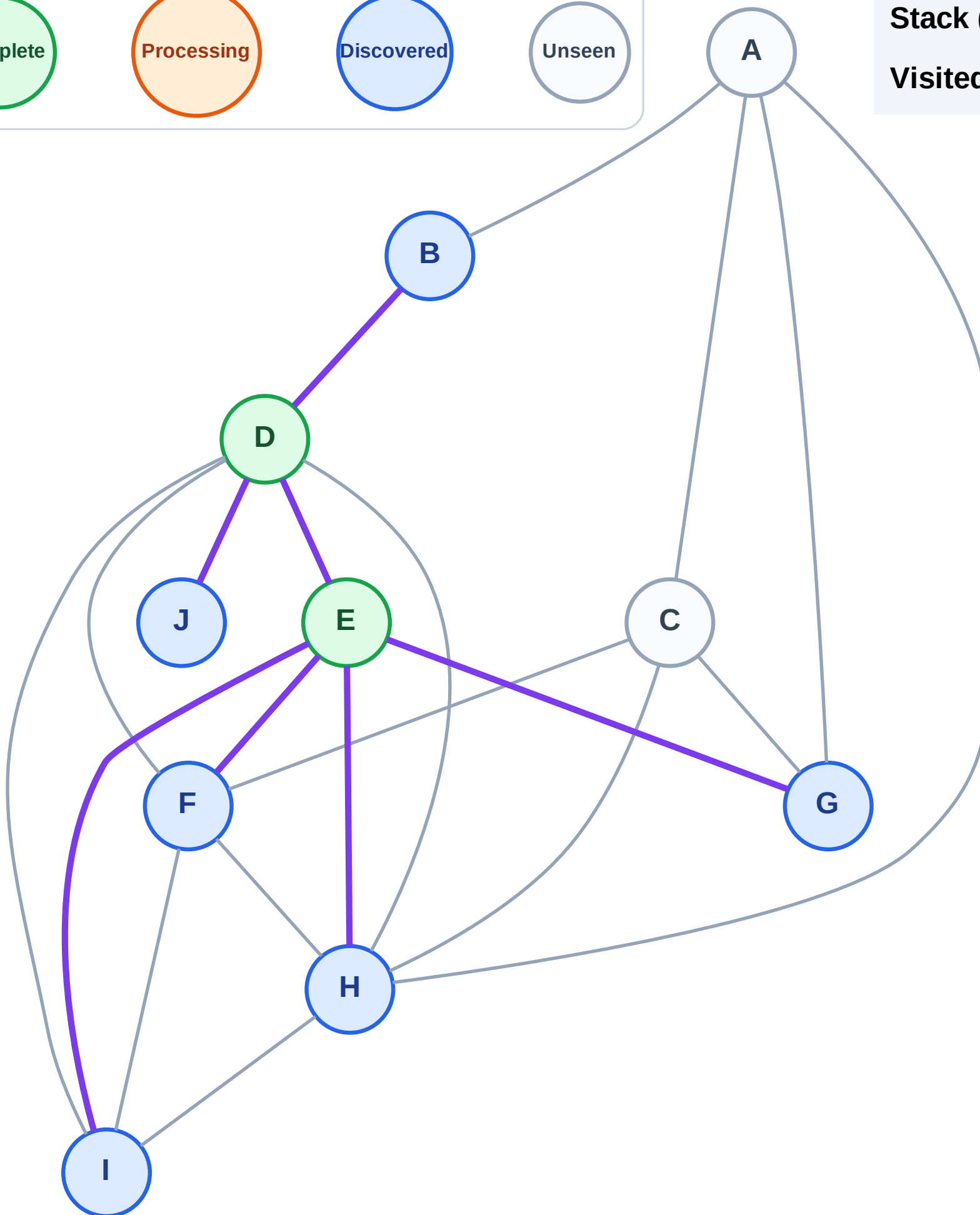
Step 5: Discover J, B from D

Legend

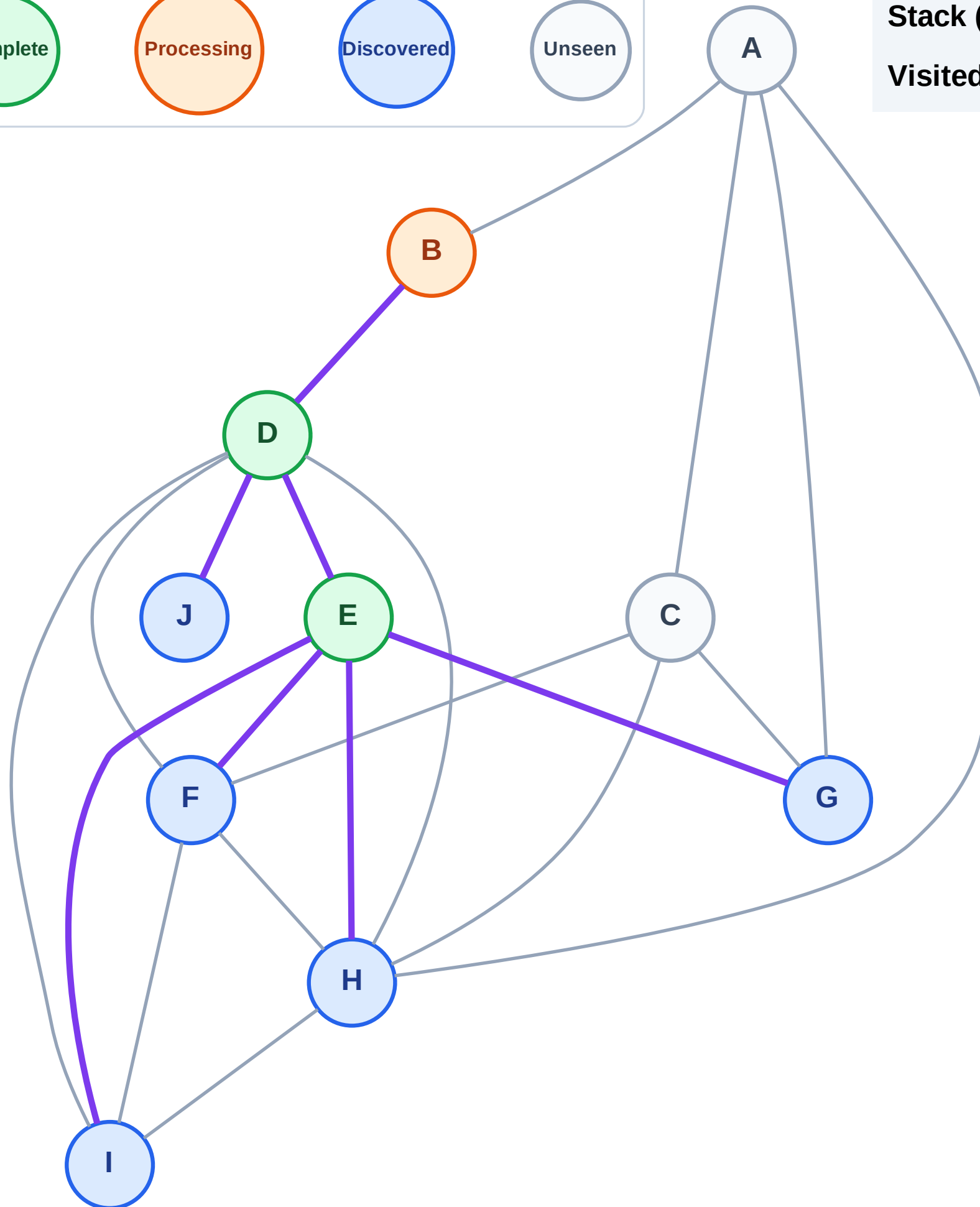
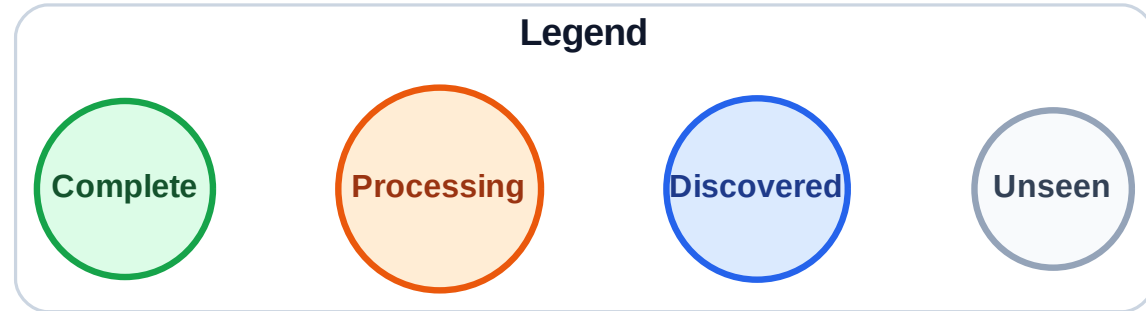
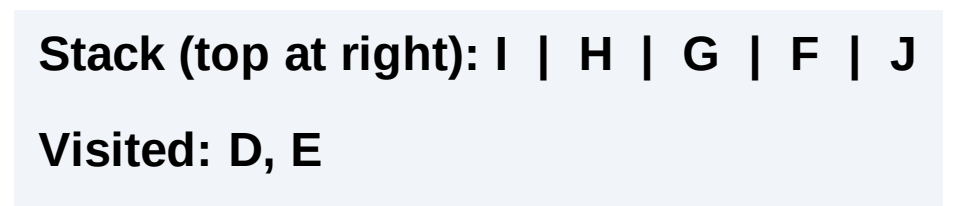


Stack (top at right): I | H | G | F | J | B

Visited: D, E



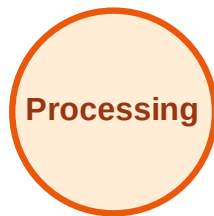
Step 6: Process node B



DFS Traversal

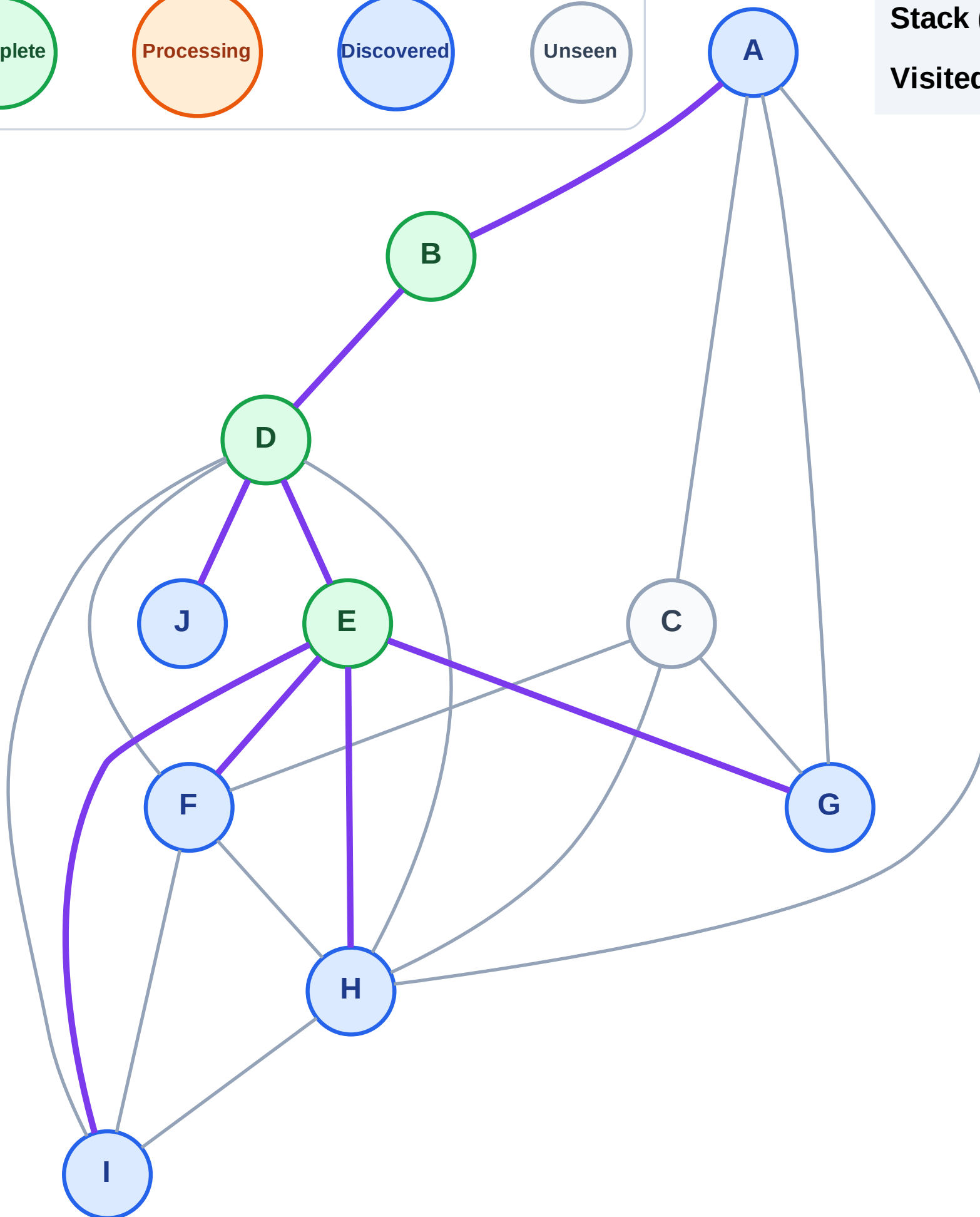
Step 7: Discover A from B

Legend



Stack (top at right): I | H | G | F | J | A

Visited: B, D, E



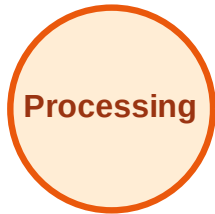
DFS Traversal

Step 8: Process node A

Legend



Complete



Processing



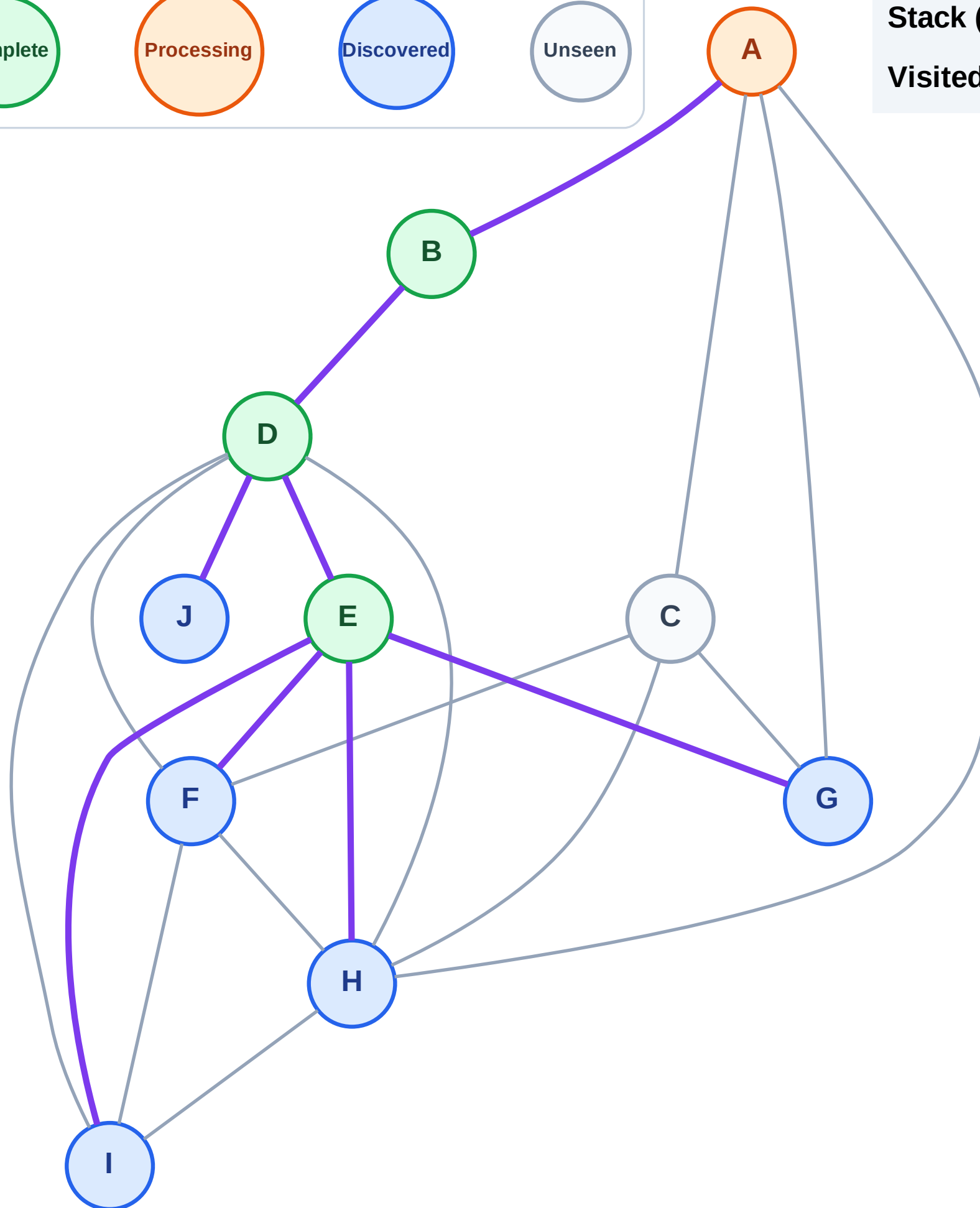
Discovered



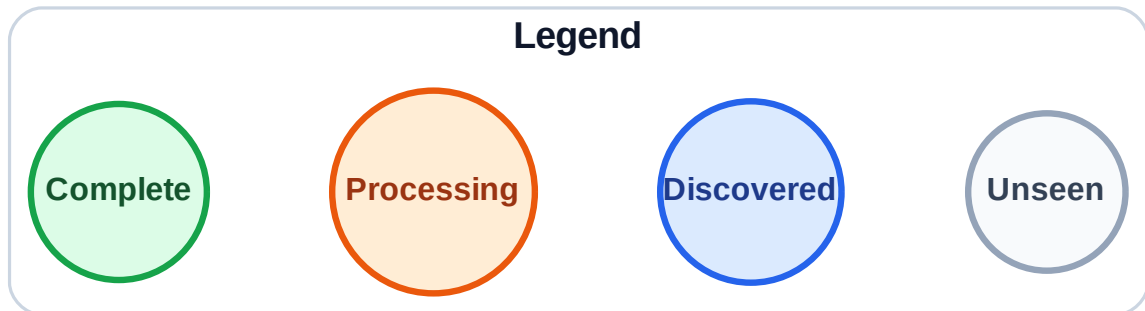
Unseen

Stack (top at right): I | H | G | F | J

Visited: B, D, E

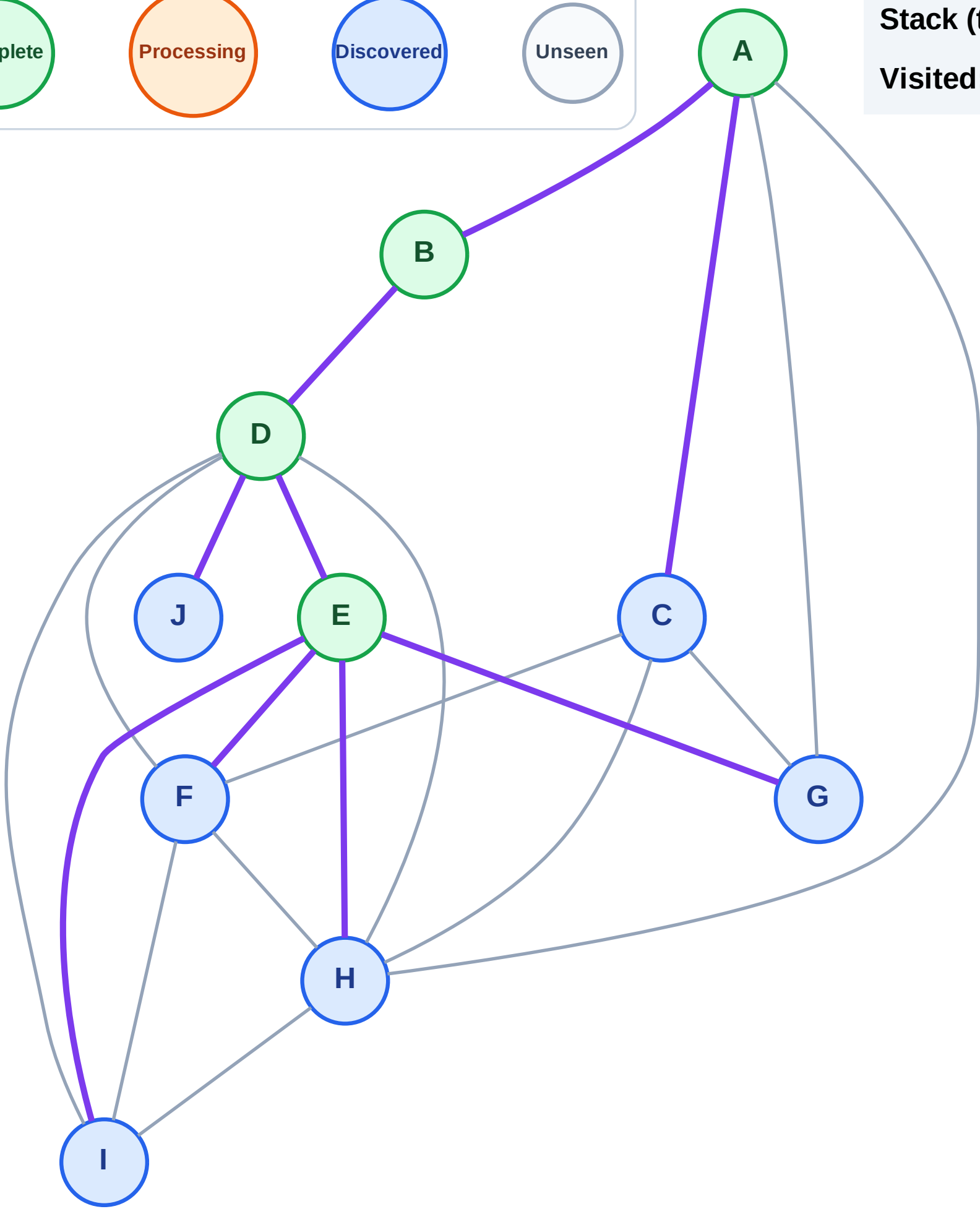


Step 9: Discover C from A



Stack (top at right): I | H | G | F | J | C

Visited: A, B, D, E



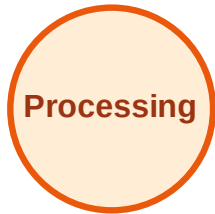
DFS Traversal

Step 10: Process node C

Legend



Complete



Processing



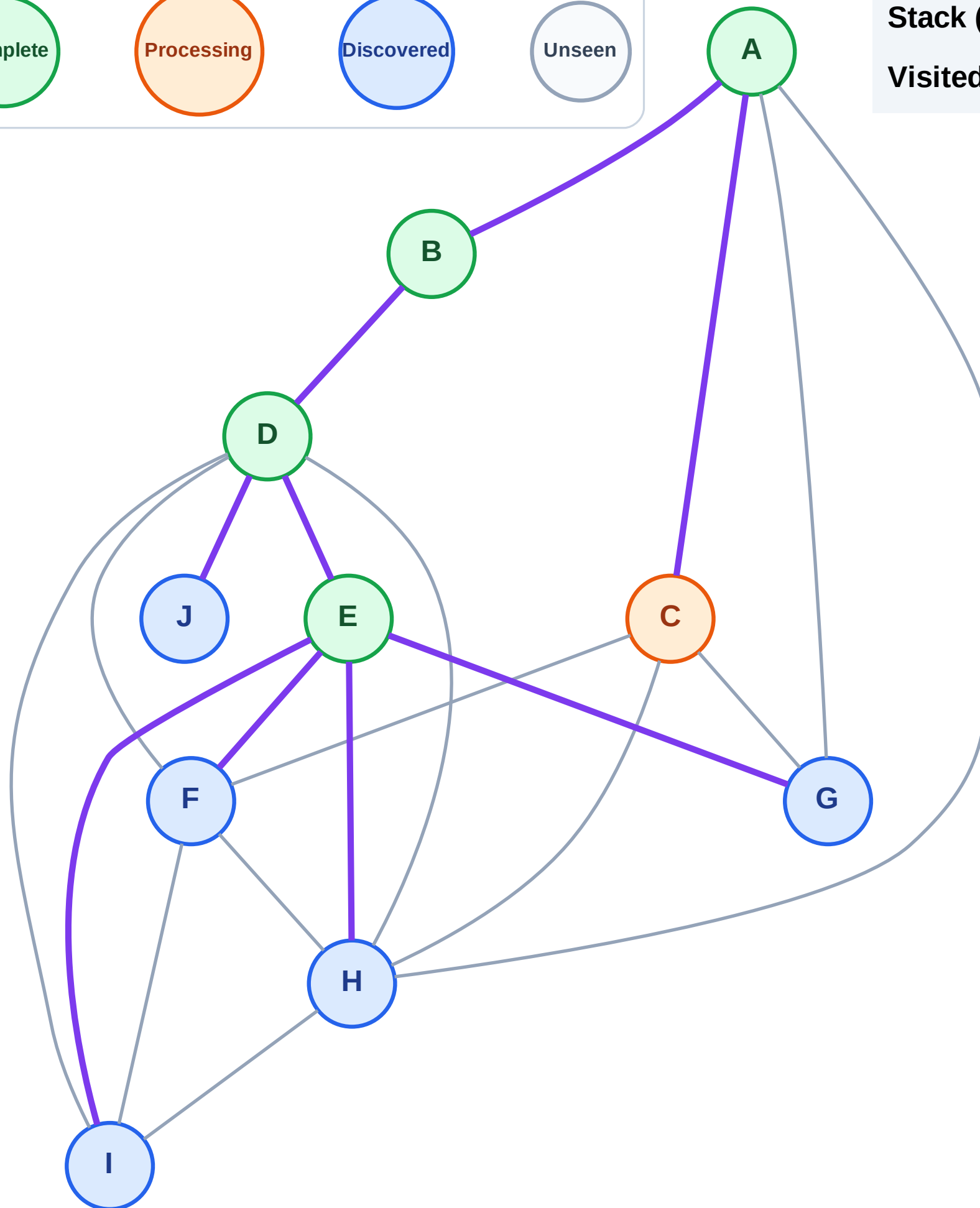
Discovered



Unseen

Stack (top at right): I | H | G | F | J

Visited: A, B, D, E

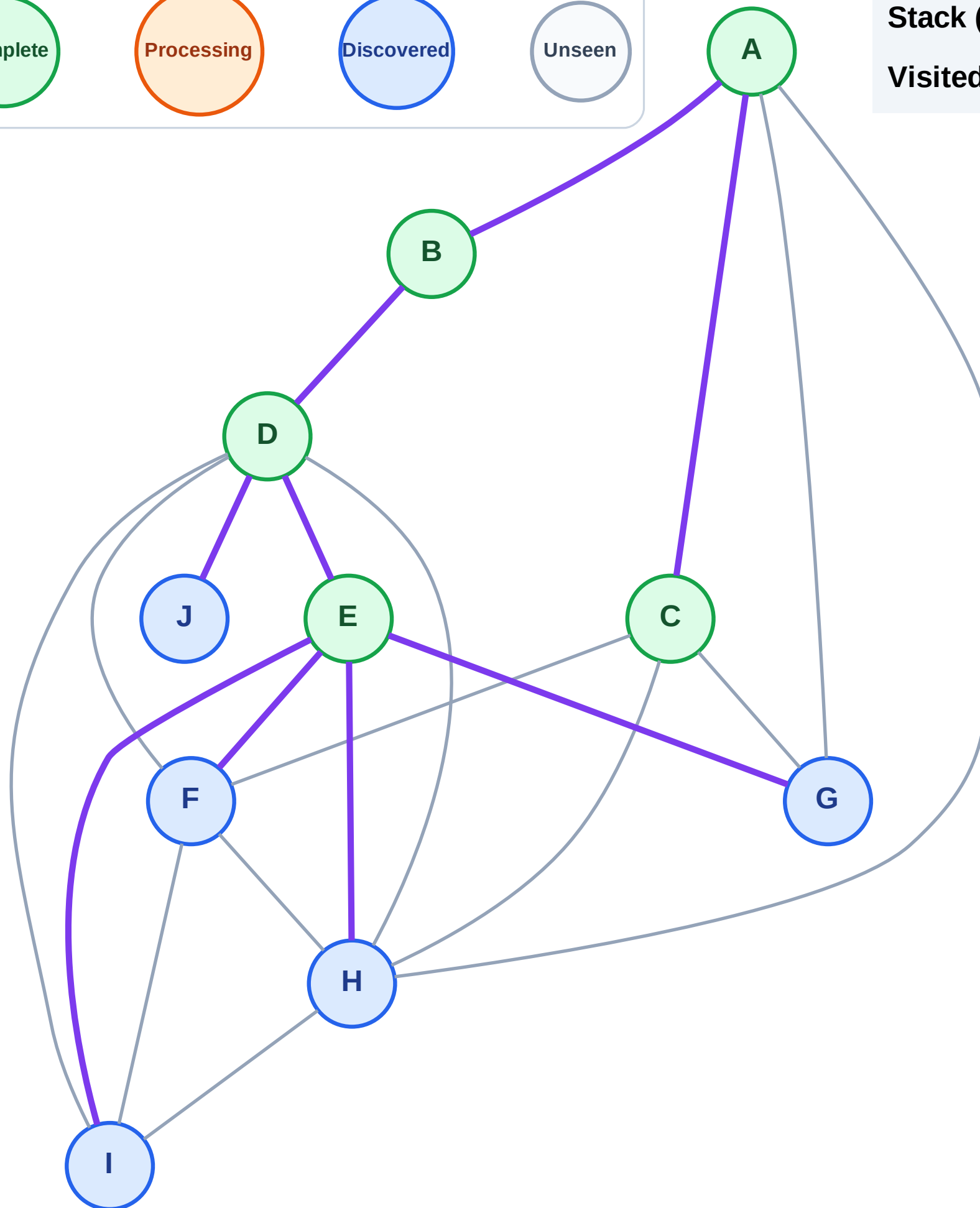


Step 11: No new neighbors from C

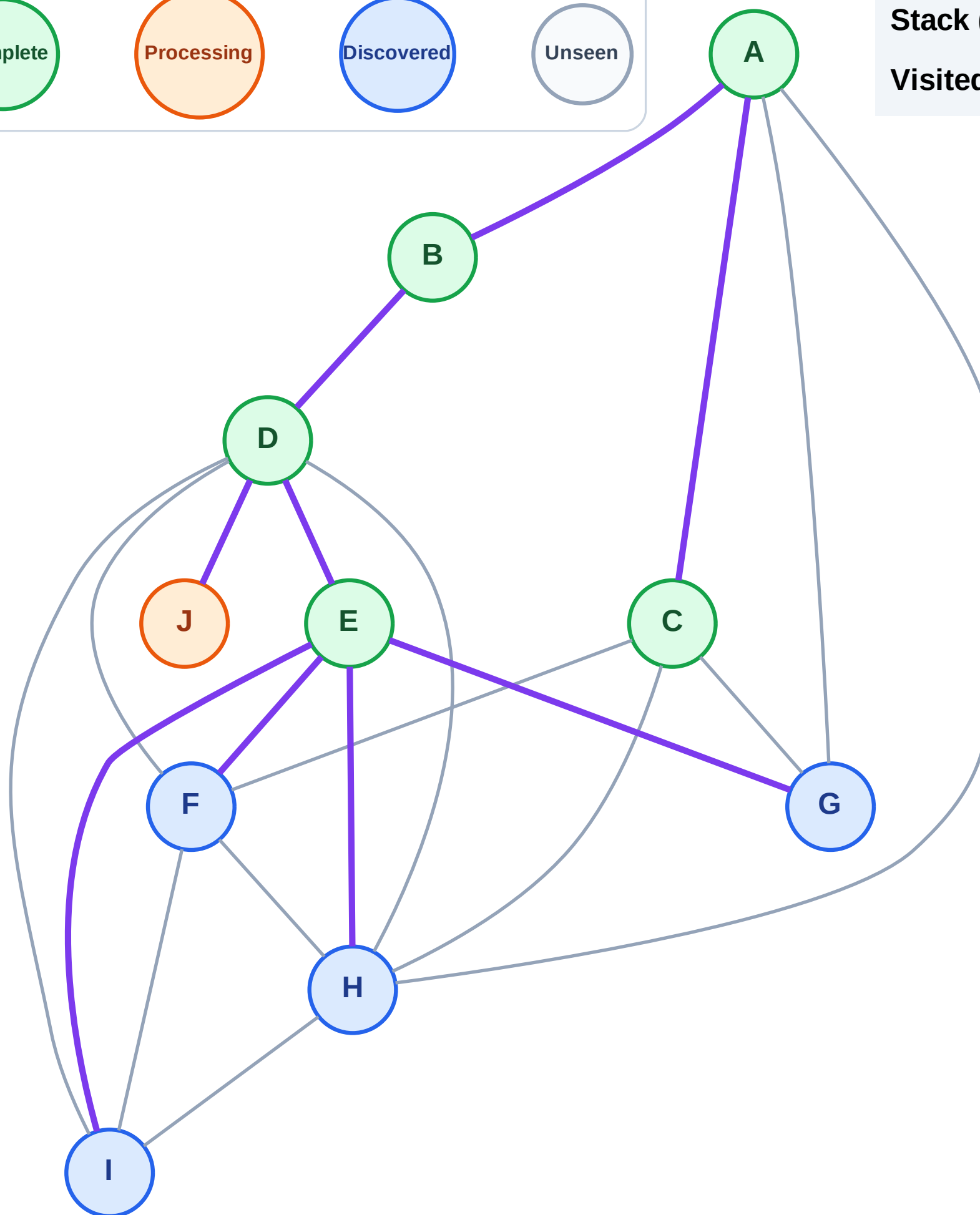
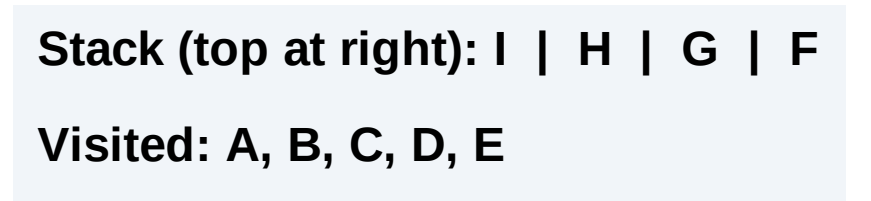
Legend



Visited: A, B, C, D, E



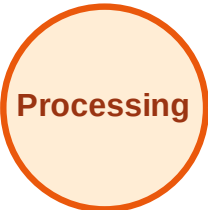
Step 12: Process node J



DFS Traversal

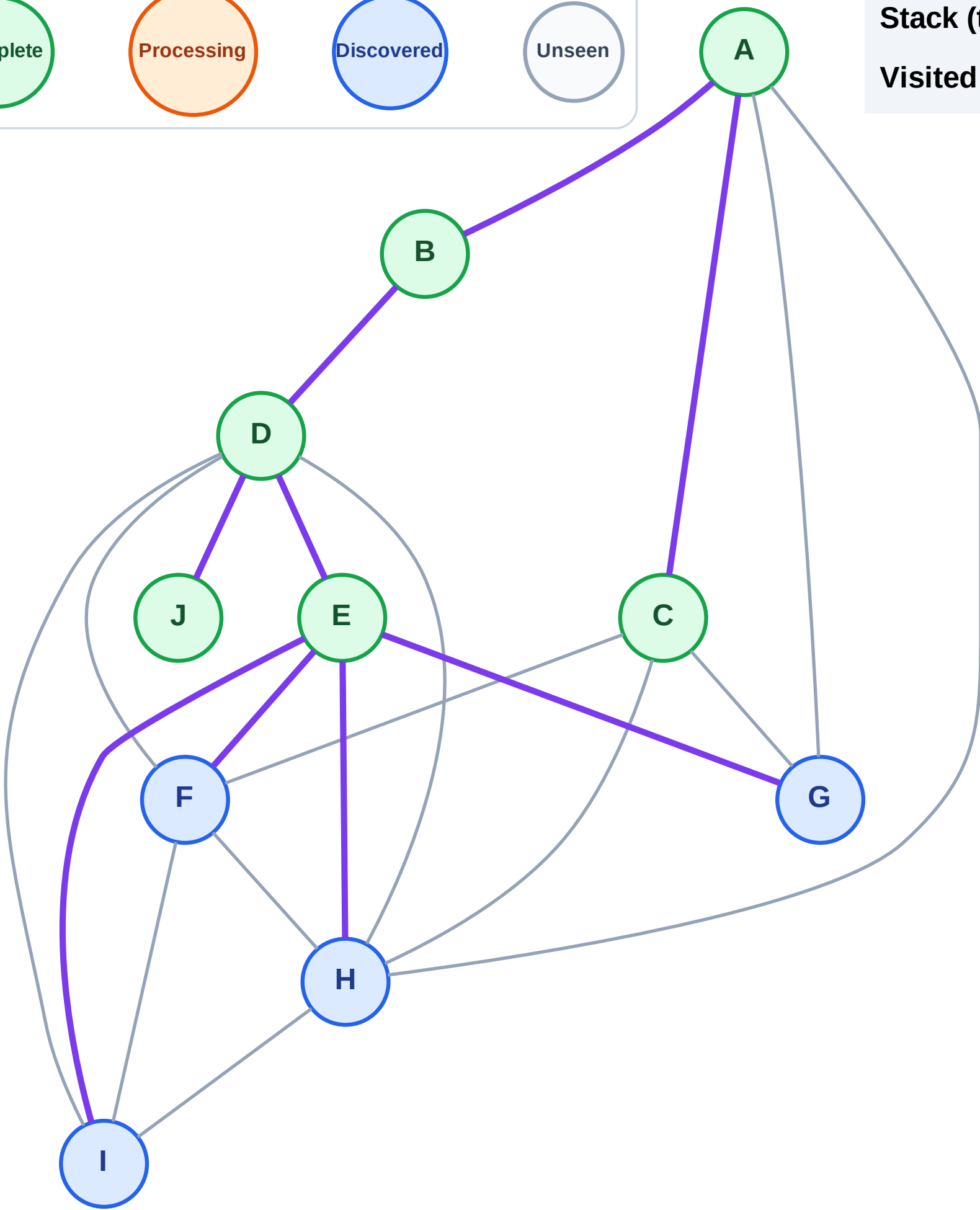
Step 13: No new neighbors from J

Legend



Stack (top at right): I | H | G | F

Visited: A, B, C, D, E, J



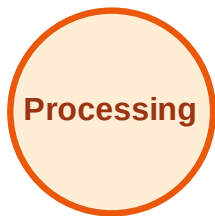
DFS Traversal

Step 14: Process node F

Legend



Complete



Processing



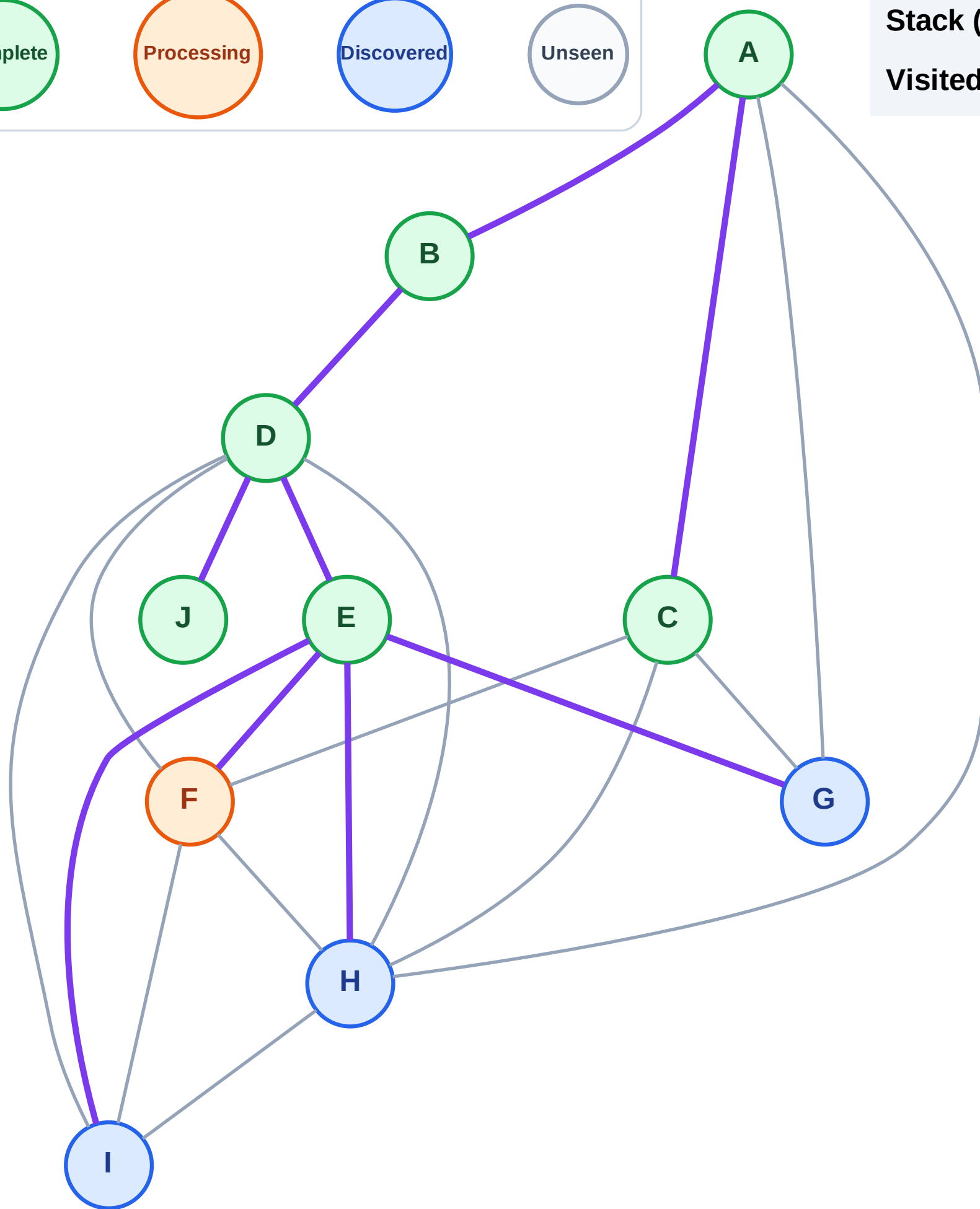
Discovered



Unseen

Stack (top at right): I | H | G

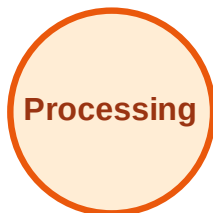
Visited: A, B, C, D, E, J



DFS Traversal

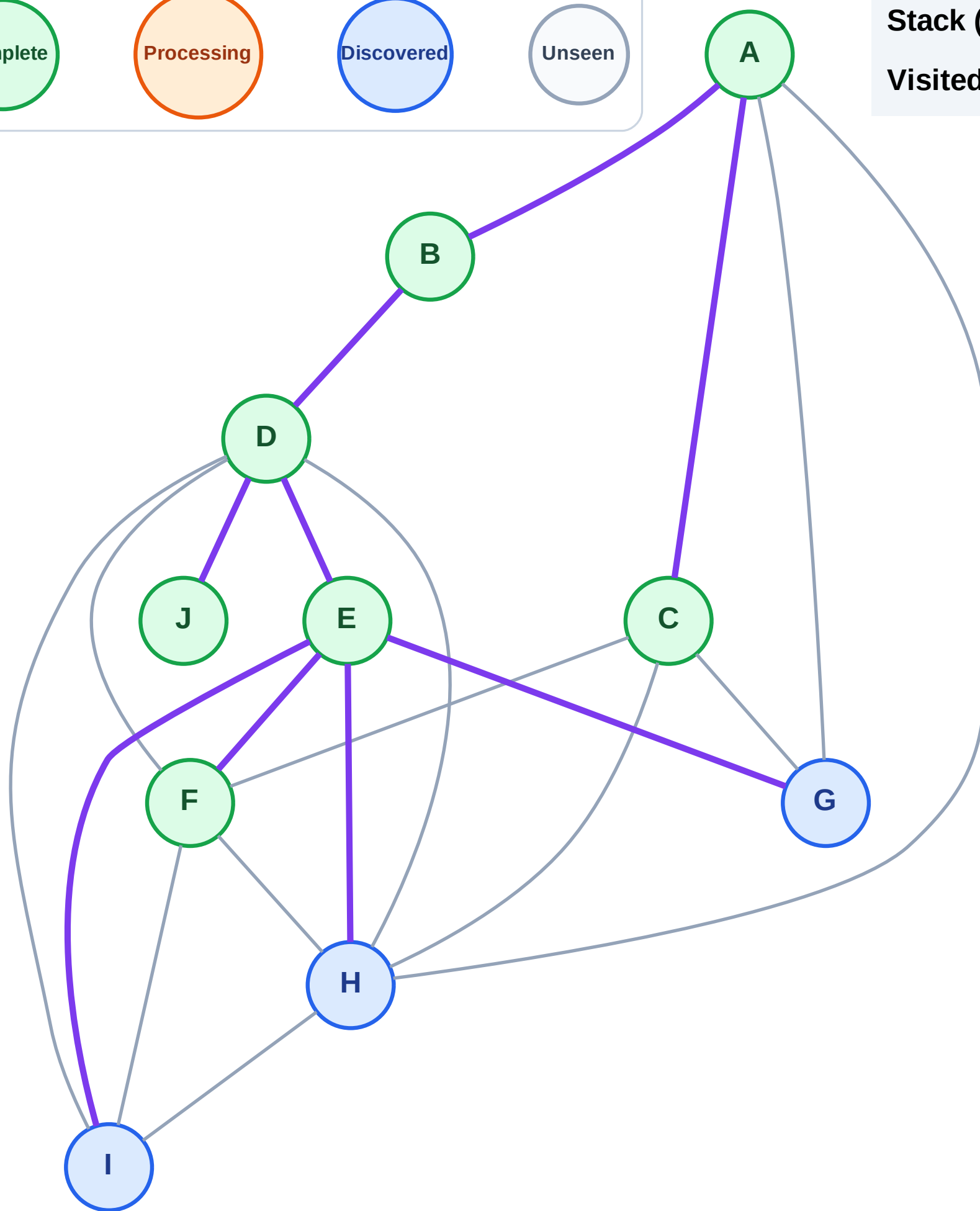
Step 15: No new neighbors from F

Legend



Stack (top at right): I | H | G

Visited: A, B, C, D, E, F, J



DFS Traversal

Step 16: Process node G

Legend

Complete

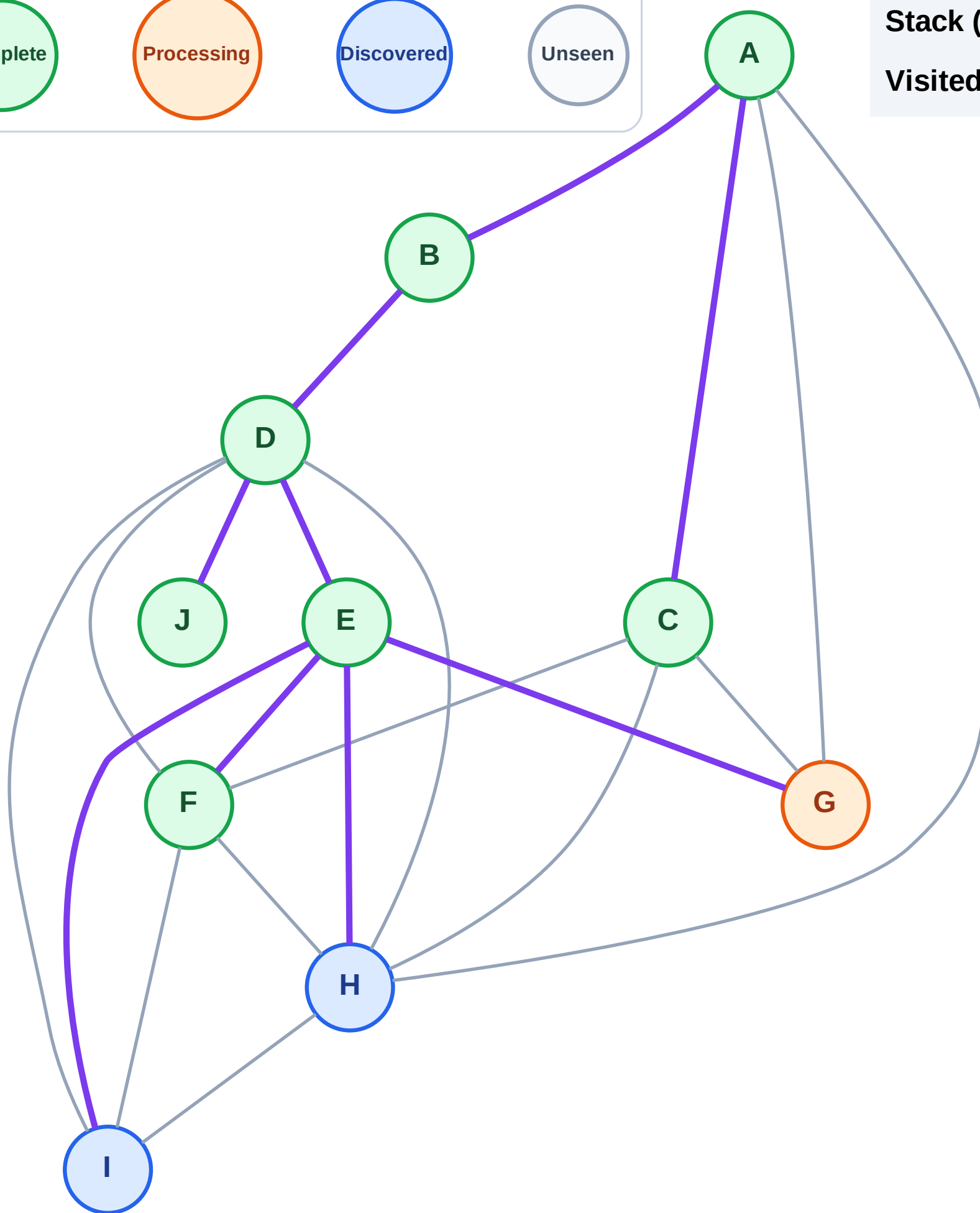
Processing

Discovered

Unseen

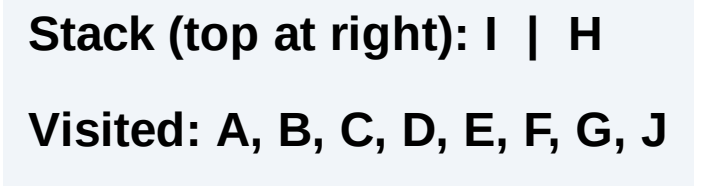
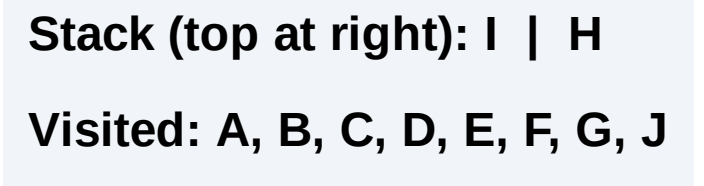
Stack (top at right): I | H

Visited: A, B, C, D, E, F, J



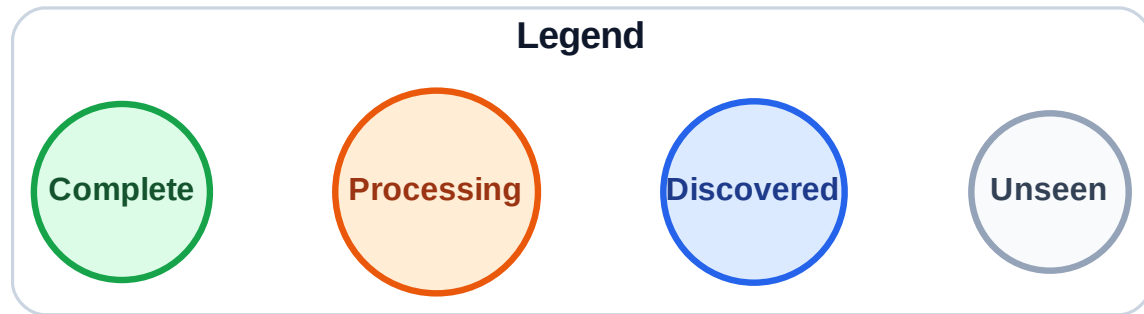
Step 17: No new neighbors from G

Legend



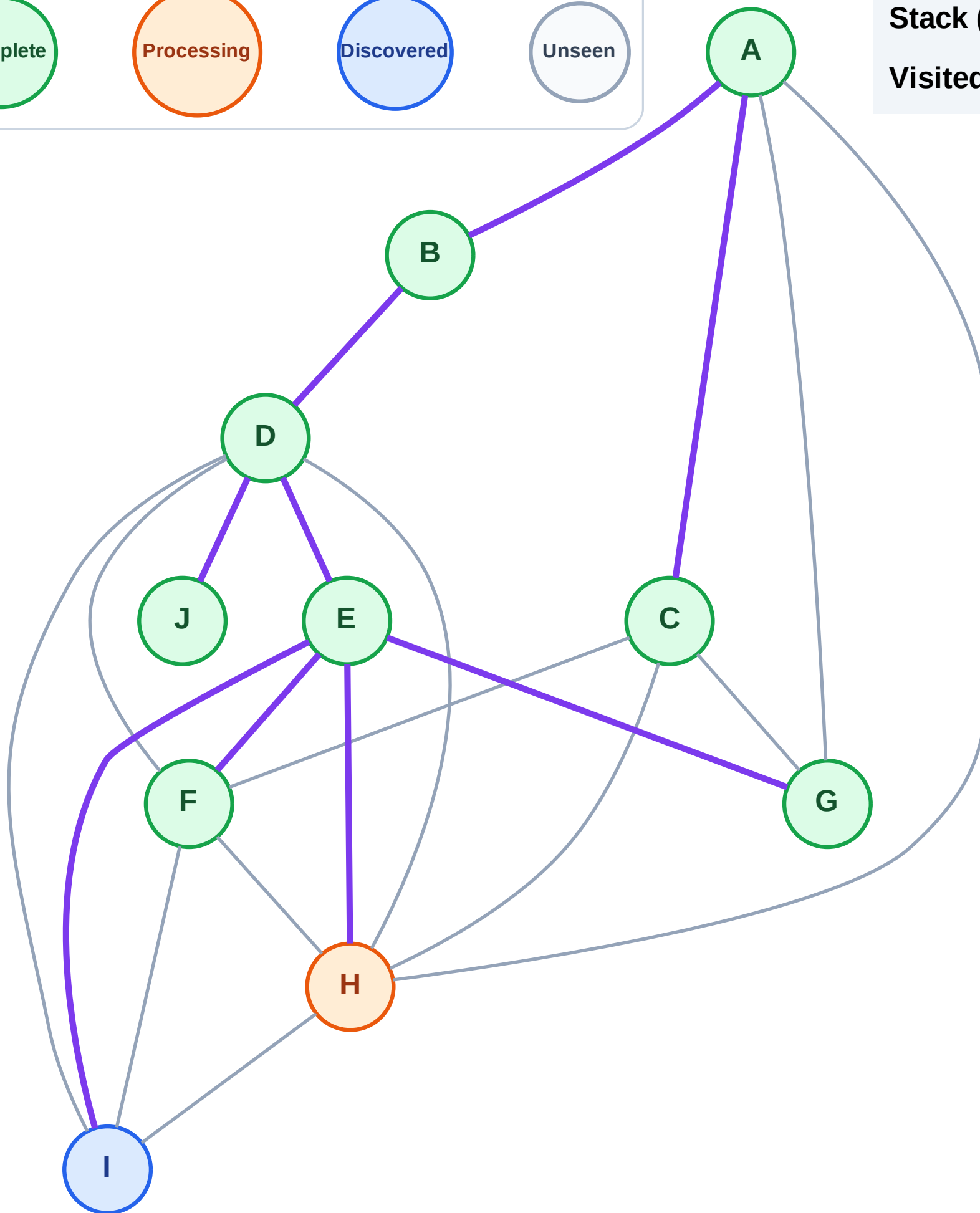
Step 18: Process node H

Step 18: Process node H



Stack (top at right): I

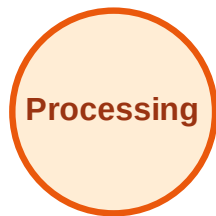
Visited: A, B, C, D, E, F, G, J



DFS Traversal

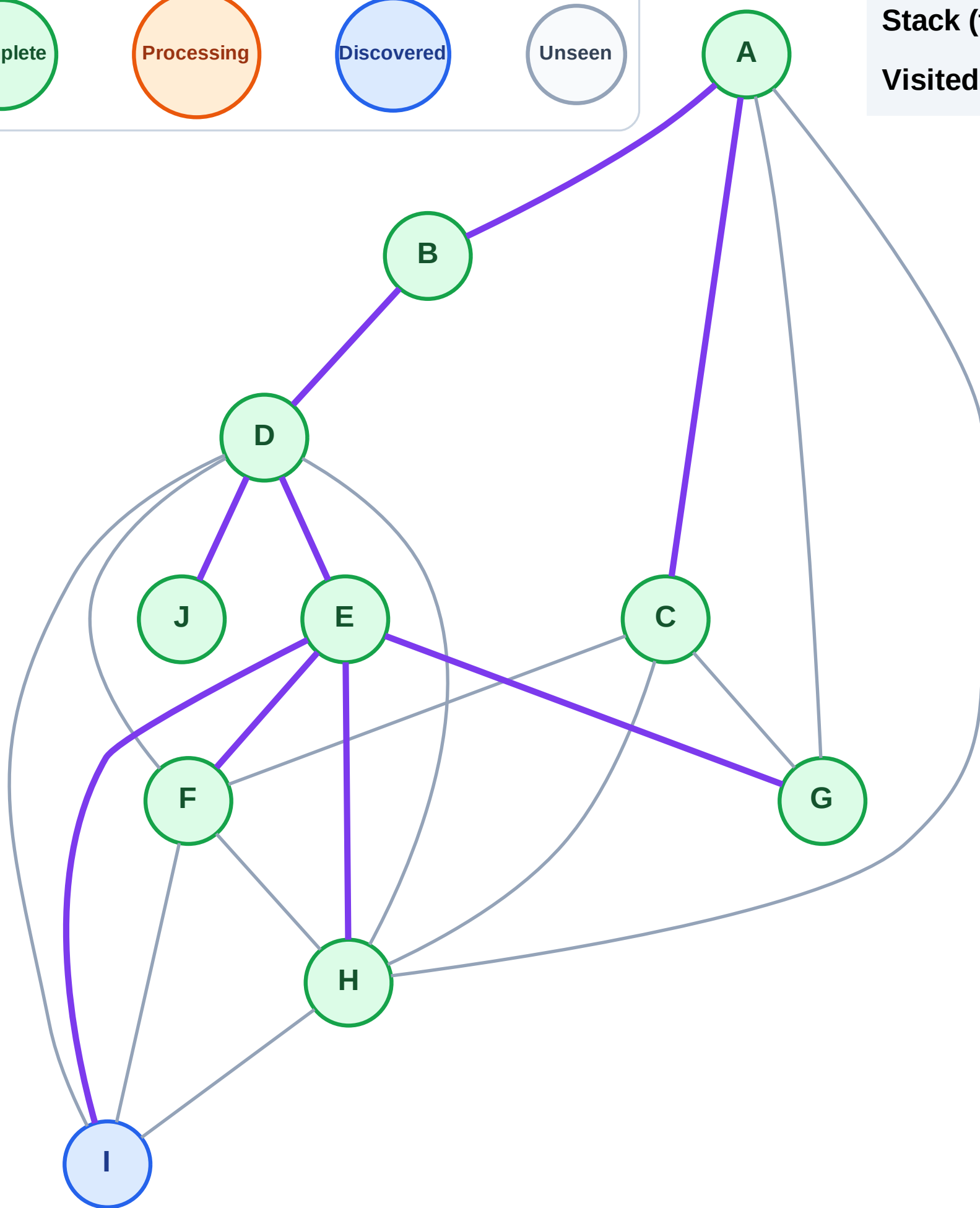
Step 19: No new neighbors from H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



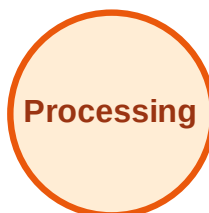
DFS Traversal

Step 20: Process node I

Legend



Complete



Processing



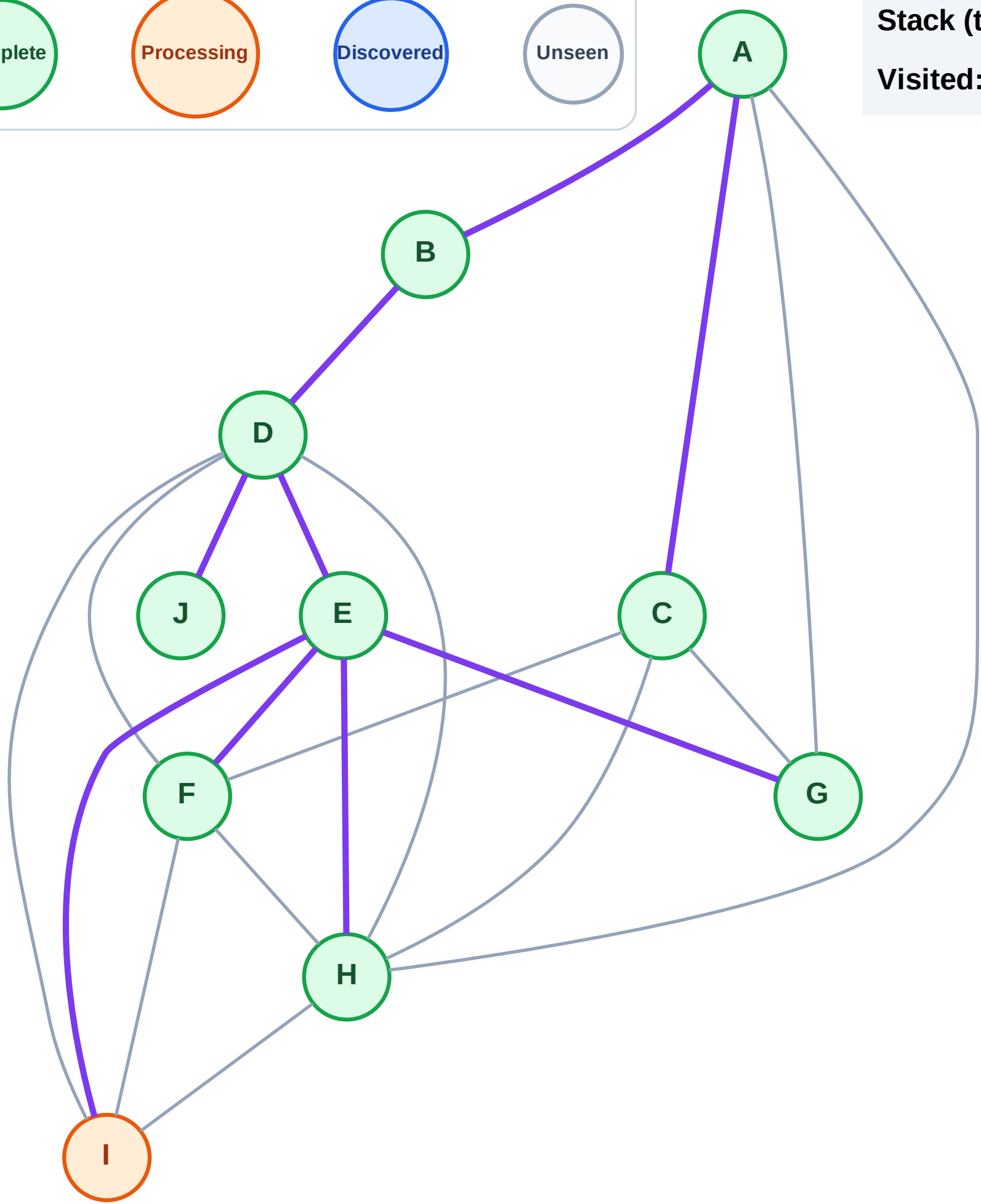
Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

